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GREENING ENERGY MARKET AND FINANCE

**FORECASTING EXTREME CLIMATE INDICES WITH MACHINE
LEARNING:**

PARAMETRIC INDEX DESIGN IN INSURANCE AND SUSTAINABLE FINANCE

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Abstract

Traditional insurance pays out only after damage is assessed, creating delays, disputes, and moral hazard. This thesis develops and statistically validates a machine-learning pipeline for forecasting extreme-event climate indices and shows how the resulting forecasts can price parametric insurance products that pay before damage occurs. The forecasting framework is the primary contribution; the insurance design is its main application.

Using the open climate index E3CL (derived from ERA5 reanalysis), I forecast six extreme-event indices for Bologna, Italy — drought, forest fire, extreme maximum temperature, extreme precipitation, extreme wind, and hail — with 16 model architectures across four horizons ($h = 1, 3, 6, 12$ months), using walk-forward cross-validation on 532 monthly observations spanning 1981–2025.

The central methodological device is the Diebold–Mariano (DM) test: only forecasts that statistically outperform the climatological benchmark are allowed to condition pricing. The parametric product's trigger is fixed at the 90th percentile; forecasts do not move the trigger but update the probability of crossing it, so premiums fall in low-risk seasons and rise in high-risk ones. This forecast-conditioned pricing on a fixed trigger is the principal operational contribution.

Of 24 hazard–horizon combinations, nine reach nominal DM significance, but only two — drought and hail at $h = 12$ — survive Bonferroni correction; the remaining seven are suggestive rather than confirmed. Skill decays with lead time, so operational design should rely on 1–6-month horizons, where forecasts can be updated and re-priced within the policy year.

The framework provides a transparent, replicable criterion for when forecast-conditioned trigger design is statistically justified, with transferability to green-finance instruments such as power purchase agreements and climate-linked bonds. Commercial deployment requires loss-data calibration, payout simulation, and actuarial premium validation.

Keywords: parametric insurance, climate index, E3CL, ERA5, machine learning, Diebold-Mariano test, trigger threshold, dynamic premium, forecast uncertainty, Bologna, Emilia-Romagna

Notation and Abbreviations

Symbol / Term	Meaning
E3CL	Open-source Extreme Climate Index Library derived from ERA5 reanalysis; provides monthly standardised anomaly indices (mean=0, std=1 over 1981–2010 baseline)
ERA5	ECMWF fifth-generation global atmospheric reanalysis (1940–present)
τ (P90)	Fixed climatological trigger: the 90th percentile of the historical index distribution ($\tau \approx +1.37$ for drought). A payout is triggered when the realised index exceeds τ . The trigger does not move with the forecast.
$\hat{\mu}_{t+h}$	Ensemble forecast mean at horizon h , issued at time t
Δ_{t+h}	Cross-model spread of the two best forecasts at horizon h ; a model-disagreement diagnostic and optional premium loading — NOT the trigger spread.
DM	Diebold–Mariano test: statistical test of whether a model significantly outperforms climatology benchmark
ACF1	Lag-1 autocorrelation coefficient; governs the R^2 -ceiling under a linear AR(1) benchmark
R^2-ceiling	Heuristic upper bound ρ^2 on R^2 achievable by an AR(1) linear predictor at $h=1$
h	Forecast horizon in months ($h \in \{1, 3, 6, 12\}$)
WII / Parametric	Weather/Climate-Index Insurance: insurance whose payout is triggered by an objective index, not assessed damage
FbF	Forecast-based financing: pre-agreed action funded when a forecast crosses a trigger threshold
σ_f	Forecast-error standard deviation at horizon h is estimated from walk-forward forecast errors. When forecast bias is negligible, RMSE serves as a practical proxy for σ_f ; if bias is material, σ_f should instead be estimated from the standard deviation of the signed errors. This quantity enters the conditional spread in $\Pr(I > \tau \mid \hat{\mu}, \sigma_f)$.
NAO / EAWR / SCAND / EPNP	Large-scale atmospheric teleconnection indices used as ML predictors
ORSA / SCR	Own Risk and Solvency Assessment / Solvency Capital Requirement (Solvency II framework)

Chapter 1: Introduction

1.1 Climate Change and the Growing Insurance Problem

Climate change is no longer a distant projection — it is an accelerating financial reality. The frequency and severity of extreme weather events have increased markedly across Europe, and particularly in the Mediterranean region, which climate projections identify as a global warming hot-spot (Giorgi & Lionello, 2008; IPCC, 2021; Seneviratne et al., 2021). In Emilia-Romagna alone, the May 2023 floods caused over €8 billion in losses. Across the continent, insured losses from natural catastrophes have increased substantially over recent decades (Munich Re, 2024; Swiss Re, 2024), while the protection gap — the share of losses not covered by insurance — remains stubbornly wide. For insurance companies, this creates a double pressure: greater liability exposure on one side and increasing difficulty pricing and reserving accurately on the other.

In Italy, these challenges are particularly acute within a rapidly evolving policy and risk landscape. Emilia-Romagna, a key agricultural region, has recently experienced compound climate shocks—including the 2023 floods and recurrent droughts—highlighting the growing exposure of agri-food systems and rural credit networks to climate volatility. At the same time, the 2024 Budget Law (Law 213/2023; Italian Republic, 2023)¹ introduces compulsory catastrophe insurance for firms, requiring coverage against major natural hazards with premiums explicitly linked to risk. This regulatory shift increases pressure on insurers to adopt transparent, statistically grounded pricing mechanisms (Neri & Prettenthaler, 2024), reinforcing the need for credible index-based approaches capable of managing basis risk and maintaining policyholder trust.

The core problem is not just that extreme events are more frequent. It is that traditional indemnity insurance — which waits for damage to occur, then pays after an assessment — is structurally misaligned with the nature of climate risk. Damage assessment is slow, costly, and contested. Moral hazard is endemic: clients have incentives to exaggerate losses (Clarke, 2016). And the very act of waiting until after damage has materialised means the insurer is always reactive rather than anticipatory. For agricultural clients, a drought payout received two months after harvest failure does not save the season.

1.2 Why Parametric Insurance? The Case for Paying Before Damage

Parametric insurance addresses these structural failures by replacing damage assessment with an objective trigger: when a pre-defined climate index exceeds a pre-agreed threshold, the payout is made automatically, regardless of actual loss (Barnett et al., 2008; Jensen & Barrett, 2017). No inspector is needed. No claim is filed. Moral hazard is structurally reduced, because the trigger is determined by objective weather data rather than self-reported damage assessments (Clarke, 2016).

The advantages are multiple. First, speed: payouts arrive when they are still useful — before or immediately after the event, when cash flow matters most. Second, transparency: the trigger rule is known in advance by both insurer and insured, eliminating disputes. Third, scalability: because no loss assessment is needed, parametric products can cover more clients at lower administrative cost. Fourth, and most importantly for this thesis: the design admits a forward-looking, forecast-based

¹ Law 30 December 2023, No. 213. Bilancio di previsione dello Stato per l'anno finanziario 2024... Gazzetta Ufficiale della Repubblica Italiana, Serie Generale No. 303, 30 December 2023.

variant. Instead of waiting for the index to exceed the threshold in real time, we can forecast the index and trigger pre-emptive action — or at least reserve capital — based on a statistically trusted forecast (Coughlan de Perez et al., 2015).

The same forecast-conditioned logic applies beyond insurance and is particularly relevant to the Greening Energy Market and Finance context of this thesis. Renewable energy power purchase agreements (PPAs) price volumetric risk using wind and temperature forecasts: a wind farm generating less power than contracted faces financial penalties, and the size of that risk depends directly on forecast uncertainty. Climate-linked green bonds embed objective index triggers into debt structures, requiring issuers to demonstrate that their index criteria are transparent and independently verifiable (ICMA, 2025). In all three cases, the missing ingredient is a statistically evaluated forecasting framework that can support objective index design in these contexts — precisely what this thesis provides. The forecasting framework is the primary contribution of this thesis; the insurance application is its main demonstration, and the same DM-confirmed forecasts are methodologically transferable to these broader green finance contexts, as discussed in Section 5.7 and the conclusions.

1.3 Climate Indices and E3CL: Objective Basis for Index Design

For parametric insurance to work, the index must be objective, freely available, and closely related to actual hazard exposure (Carter et al., 2014; Conradt et al., 2015). This thesis uses the E3CL (Extreme Climate Index Library), an open-source dataset derived from ERA5 reanalysis data provided by ECMWF (ECMWF, 2023). E3CL provides monthly time series of six extreme-event indices for Bologna (BO) and other European stations. All indices are expressed as monthly standardised anomalies (mean 0, standard deviation 1 over the 1981–2010 baseline), making them directly interpretable: an index value of +2.0 means the variable is 2 standard deviations above the historical mean.

The six variables studied are: Drought Index, Forest Fire Frequency, Extreme Maximum Temperature, Extreme Precipitation, Extreme Wind Speed, and Hail Frequency. Together, these cover the main climate hazards relevant to agricultural and property insurance. ERA5-derived indices have been validated against station observations across Europe and show excellent agreement for temperature and drought at Bologna's location; ERA5 reanalysis fields more generally have been the subject of detailed evaluation studies (Lavers et al., 2022). Having an objective, openly verifiable index in place is only half of the design problem, however; the other half is how the trigger built on that index is set — which is where static and forecast-informed designs diverge.

Traditional parametric insurance uses a static threshold calibrated from historical data alone — for example, ‘pay if the drought index exceeds a fixed level in any month of the season.’ A static design also charges, in effect, the same premium every year. In low-risk seasons the policyholder overpays; in high-risk seasons the price understates the true risk. This misalignment deters low-risk farmers, worsens affordability, and aggravates adverse selection, because those most likely to buy are precisely those who expect a bad year.

A forecast-informed design keeps the contractual trigger fixed at a transparent climatological return-period level but updates the conditional probability of crossing it — and hence the premium — using a statistically validated seasonal forecast (Conradt et al., 2015; Figueiredo et al., 2018). In a year when the drought forecast points strongly positive, the conditional crossing probability rises, the insurer reserves more capital, and the premium is higher; in a benign year the premium falls, making the

product more competitive and affordable. The trigger itself never moves, so the contract remains transparent and auditable; what adapts is the forecast-conditioned price.

It is worth being explicit about the division of labour, because it defines the contribution of this thesis. The trigger defines what counts as an extreme climate event; the forecasting model does not redefine the event but estimates its probability of occurrence this season, which is what drives the premium and the reserve. The model does not decide what risk is — it decides how likely that risk is this season. The forecast-informed pricing mechanism is therefore the principal contribution; the fixed trigger is the setting that demonstrates it, and an adaptive, loss-data-driven trigger is identified as the natural next step (Section 6.4).

1.4 Structure of the Thesis and Research Questions

This thesis pursues one overarching question: Can ML-based climate forecasting be statistically validated to a level of rigour that supports its use as an evidential basis for parametric index design in insurance and green finance? From this, three specific research questions follow:

RQ1 — Forecast ability: For each of the six E3CL climate variables at Bologna, which model families achieve statistically significant improvement over climatology (as measured by the DM test) across forecast horizons of 1, 3, 6, and 12 months?

RQ2 — Threshold Design: How can DM-validated forecasts be used to estimate the conditional probability of exceeding a fixed climatological trigger and thereby inform pricing?

RQ3 — Insurance Implications: What do the forecast results imply about the design and pricing of parametric products for each climate variable? Which variables and horizons support dynamic products versus static ones?

This thesis makes one principal operational contribution: a forecast-informed pricing mechanism in which DM-validated seasonal forecasts shift the conditional probability of exceeding a fixed climatological trigger and thereby dynamically adjust the insurance premium, while the trigger itself remains transparent and static. The fixed climatological trigger is the transparent, auditable contract structure adopted to demonstrate this mechanism; it is not in itself the contribution. The economic rationale is to align the premium with seasonal risk, improving affordability and mitigating adverse selection. Two further ideas are introduced as conceptual extensions and directions for future work, and are explicitly not part of the operational pricing framework — the principal barrier in both cases being the absence of granular insurance loss data: (i) a data-driven adaptive (moving) trigger, in which the payout threshold itself is learned from realised loss data using actuarial or machine-learning methods, replacing the fixed trigger; and (ii) a hybrid layered architecture combining anticipatory, parametric, and residual-indemnity payouts (both in Section 6.4).

The study is geographically limited to the Bologna (BO) station in Emilia-Romagna, northern Italy. The temporal scope is January 1981–April 2025 (532 monthly observations), with 12 months of out-of-sample hold-out (May 2025–April 2026) used exclusively for forecast accuracy visualisation. The analysis is at monthly resolution, chosen for its relevance to seasonal insurance products. No actual loss data are used; the thesis focuses on hazard forecasting and trigger design. The insurance design examples are illustrative, using a stylised €1 million notional consistent with agricultural cooperative scales. All empirical results are therefore strictly station-specific: they characterise the behaviour of E3CL indices at Bologna under the chosen sample and do not by themselves justify extrapolation to other regions or crops without replication. From an operational standpoint, insurers do not need a

perfectly accurate 12-month forecast to design viable parametric covers: what matters is whether they can obtain statistically credible information at decision horizons that align with underwriting and capital-planning cycles (typically 1–6 months ahead). This thesis therefore emphasises short- and medium-horizon forecasts for trigger and premium updates and treats annual-horizon results as a stress-test of the upper limits of what current models can achieve rather than as a direct input to pricing.

The remainder of this thesis is organized as follows. Chapter 2 reviews the literature on parametric insurance, forecast-based financing, ML applications in climate-risk assessment, green finance instruments, and climate triggers, while positioning the proposed forecast-conditioned pricing framework. Chapter 3 describes the E3CL and ERA5 datasets, diagnostic procedures, forecasting methodology, validation design, Diebold–Mariano testing, and the fixed-trigger, forecast-conditioned premium framework. Chapter 4 presents empirical results, interpretation, limitations, and hold-out validation. Chapter 5 translates the findings into financial applications — parametric insurance pricing recommendations, premium sensitivity analyses, Solvency II implications, and green finance applications. Chapter 6 concludes and outlines future research directions. Appendices provide complete DM and RMSE tables, supplementary figures, and premium sensitivity calculations.

Figure 1.1 summarises the complete conceptual framework of this thesis as a single logic chain and is best read as a map of the chapters that follow. As Figure 1.1 shows, climate hazard data (ERA5/E3CL) flows through signal diagnostics, the multi-model forecasting pipeline, the Diebold–Mariano (DM) statistical gate, the Δ uncertainty diagnostic, and the forecast-conditioned pricing step on a fixed climatological trigger, arriving finally at parametric index design and ORSA implications. This chain — set out visually in Figure 1.1 — is the core methodological contribution of the thesis: each arrow represents a transformation that is evidentially grounded and statistically gated. The Diebold–Mariano test (Diebold & Mariano, 1995), introduced informally here and defined in full in Section 3.4, is the gate that admits only forecasts with statistically significant skill over a climatological benchmark into the pricing layer.

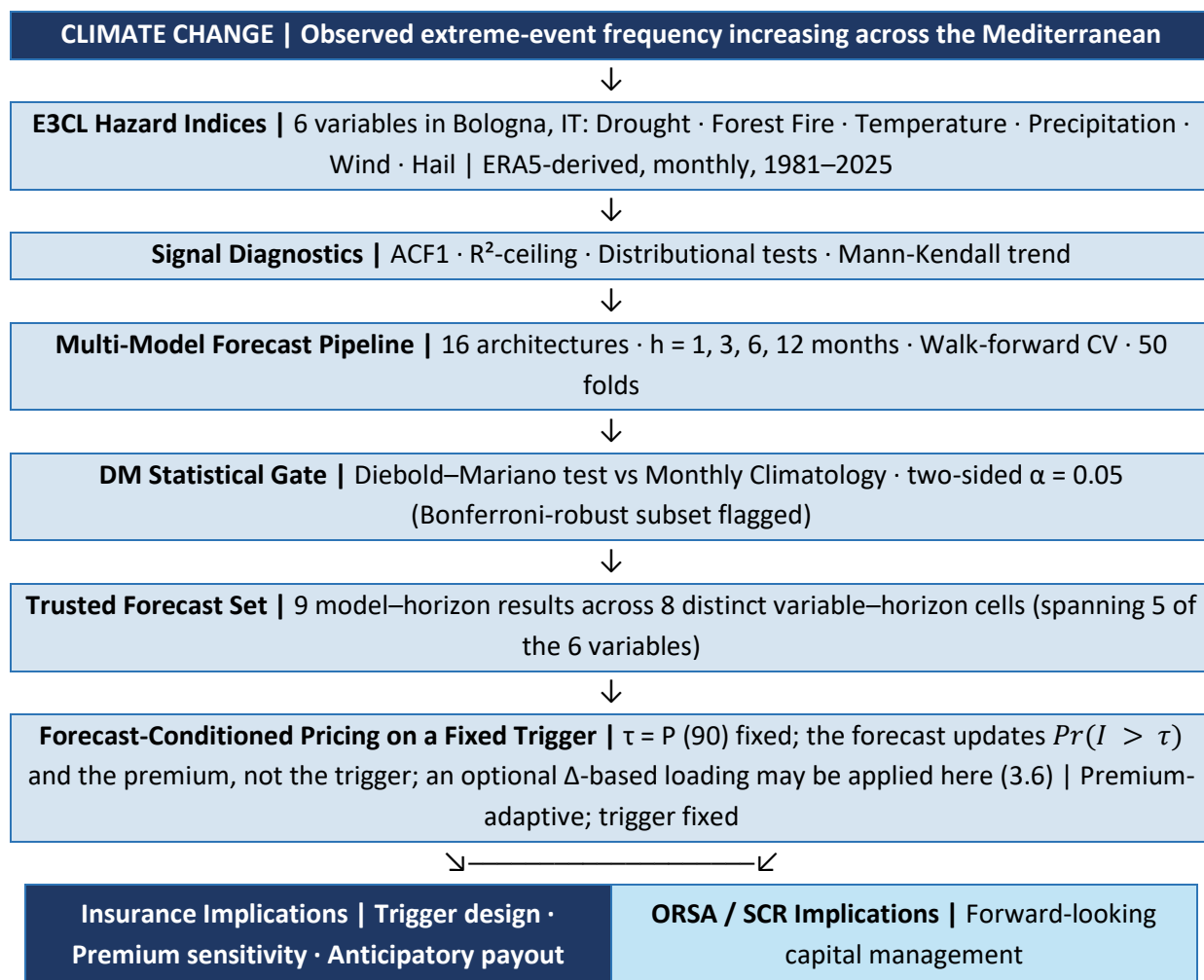


Figure 1.1. High-level architecture of the thesis workflow from climate data to insurance design. Each step is supported by statistical evidence; all insurance implications are conditional on a stable index–loss relationship.

Chapter 2: Literature Review

2.1 Parametric Insurance: Mechanisms, Benefits, and Basis Risk

The literature on parametric insurance has grown rapidly alongside climate finance. The fundamental trade-off, documented by Clarke (2016), is between basis risk and speed: parametric products pay quickly and cheaply, but may diverge from actual losses. Jensen & Barrett (2017) show that the welfare effects of parametric products are positive only when the index-loss correlation is high enough; below a critical R^2 threshold, the product can harm purchasers who receive payouts when they have no losses, or vice versa. Figueiredo et al. (2018) extend this to show that probabilistic trigger design — where the trigger is a conditional exceedance probability rather than a point value — can substantially reduce basis risk.

This thesis operates in the 'hazard module' of this framework: it builds the best possible forecast of the climate index, leaving the index-to-loss calibration for future work when ISTAT agricultural loss data for Emilia-Romagna become available. The relevant benchmark from the European literature is an index-to-loss R^2 in the range 0.3–0.6 for comparable studies (Prettenhaler et al., 2012; Eltazarov et al., 2023), implying residual basis risk of 40–70%. It is important to stress that these index-to-loss R^2 ranges are drawn from studies in different countries, crops, and index constructions, and no empirical basis-risk estimate currently exists for E3CL-derived drought indices in Emilia-Romagna. They are cited here only as illustrative benchmarks for how large the index–loss correlation would need to be for welfare results such as those in Clarke (2016) to become relevant.

2.2 Climate Risk and Insurance Under Solvency II: From Static to Dynamic Trigger Design

Most operational parametric products use static triggers calibrated from historical distributions. The emerging literature on forecast-based financing (FbF), developed by humanitarian organisations including the Red Cross (Coughlan de Perez et al., 2015; Zhu et al., 2025), demonstrates that this design can be improved by replacing the real-time index observation with a forecast. If the forecast predicts that the index will exceed the trigger threshold before the event occurs, action (or pre-payment) can be initiated earlier. The DM test logic in this thesis directly mirrors the activation-threshold logic in FbF: only forecasts with statistically confirmed skill are used to trigger action.

The operational FbF literature has expanded rapidly since the early Red Cross pilots, with detailed trigger-calibration protocols now available from the Red Cross Red Crescent Climate Centre, IFRC, and related initiatives. These guidelines typically combine expert judgement, historical verification, and simple skill metrics, but they rarely apply formal DM-type tests across a broad set of competing models. Similarly, recent weather-index insurance studies in Latin America and other emerging markets increasingly use machine-learning forecasts (e.g. Random Forests and LSTMs) to build rainfall or runoff indices, yet focus primarily on adoption decisions, subsidy design, and farmer welfare rather than on systematic, multi-architecture forecast evaluation (Bedoya Soto et al., 2025). The contribution of this thesis is therefore not to claim that FbF and index-insurance designs ignore forecast skill altogether, but to propose a transparent, DM-gated evaluation layer that makes the strength of the forecasting evidence explicit and reproducible across many models and horizons.

For European insurance companies, the shift toward forecast-driven products also has capital implications. Under Solvency II, the Solvency Capital Requirement (SCR) is calibrated to a 99.5th

percentile loss over one year. If a company holds a portfolio of parametric climate contracts, the SCR should reflect the conditional probability of trigger events. A static calibration based on historical mean frequency does not capture the dynamic risk landscape; a forecast-conditioned calibration does. The Geneva Association (2024) identifies parametric index-based covers as particularly suitable for climate risk adaptation, and EIOPA (2023) guidance on climate-related ORSA explicitly supports the use of forward-looking models with documented statistical validation.

This creates a direct regulatory motivation for the DM-test-based approach in this thesis: an insurer that can show a regulator a DM-validated forecast is in a stronger position to justify dynamic SCR adjustments than one relying on narrative claims about model performance.

2.3 Machine Learning for Climate Index Forecasting

The application of ML to climate forecasting has accelerated dramatically since 2020. Joseph et al. (2026, early online) document a clear trend toward hybrid statistical-AI approaches in weather index insurance. Alonso-Robisco et al. (2024) identify climate-data analytics as one of the fastest-growing ML application domains in finance. For the specific problem of monthly extreme-event index forecasting, however, the literature is sparse: most studies focus on sub-daily or daily resolution, or on continuous variables (temperature, precipitation totals) rather than extreme event indices. This thesis provides an applied case study that combines DM tests, E3CL-type monthly extreme indices, and parametric trigger design in a single framework, with particular attention to the conditions under which ML adds genuine value (high-signal series, appropriate horizons) versus when it fails (near-white-noise series, long horizons).

A common gap running through all three strands of this literature — parametric insurance economics, forecast-based financing, and ML climate forecasting — is the absence of a unified evaluation framework connecting statistical forecast skill to insurance trigger design via a formal, reproducible criterion. Existing parametric insurance studies typically assume static hazard distributions or test single trigger rules without assessing whether underlying forecasts are statistically superior to naïve benchmarks in a transparent, reproducible way (Carter et al., 2014; Figueiredo et al., 2018; Conradt et al., 2015). Forecast-based financing studies validate activation rules operationally but rarely apply formal forecast evaluation tests across multiple model architectures (Coughlan de Perez et al., 2015; Zhu et al., 2025). ML climate forecasting papers demonstrate predictive accuracy but generally do not connect results to insurance contract design (Alonso-Robisco et al., 2024). This thesis addresses the intersection of all three strands by applying systematic DM-based forecast evaluation across 16 architectures and 6 variables, and explicitly linking the resulting evidence tiers to trigger design recommendations — providing the methodological bridge that existing work has not yet established.

A minor, discretionary extension of this design is to let the disagreement between the two best DM-confirmed models — the cross-model forecast spread Δ — act as an optional loading on the premium when those models diverge. The idea that ensemble dispersion is informative about forecast uncertainty is well established (Bates & Granger, 1969; Timmermann, 2006); here it is used only as a discretionary underwriting adjustment, never as an input to the contractual trigger, and it is not among the validated contributions of this thesis. Its formal form is given as a conceptual extension in Section 6.4.1.

2.4 Green Finance and Climate Triggers

Green finance instruments increasingly depend on objective and verifiable climate metrics. Under the EU Green Bond Standard (EU GBS) and the ICMA Green Bond Principles (2025), issuers must provide transparent disclosure and credible, reproducible criteria for what qualifies as green, which makes the choice of trigger mechanism a material design issue. In practice, climate-linked securities and sustainability-linked structures rely on climate variables that can be independently observed and audited, making open-source indices such as E3CL, derived from ERA5 reanalysis, especially well-suited for trigger design. ERA5-based indices are freely accessible, globally maintained by ECMWF, and free from data-provider dependency — characteristics that align directly with regulatory requirements for transparency and replicability under the EU GBS.

The same logic applies to renewable energy power purchase agreements (PPAs). These long-term contracts between generators and buyers expose both parties to volumetric risk: the amount of electricity generated depends on weather conditions, particularly wind speed and temperature. Forecasts of these variables therefore have direct pricing relevance, since they improve estimates of expected output and reduce uncertainty about contract settlement. Tranberg et al. (2020) document the volumetric risk structure of long-term PPAs and show that better wind forecasts materially reduce contract pricing uncertainty. Thomaidis et al. (2023) extend this to multi-dimensional wind energy risk, demonstrating that spatial and temporal forecast quality jointly determine risk premium requirements. Both studies underscore the same conclusion: statistically rigorous, formally evaluated forecasts are operationally necessary for green finance applications.

The integration of DM-validated ML forecasts into climate-trigger design appears limited in the existing literature. Existing work on green bonds, climate-linked debt, and renewable energy contracts typically assumes static thresholds, historical calibration, or meteorological forecasts without a formal test of forecast skill relative to a reproducible benchmark. This thesis fills that gap by demonstrating how DM-validated machine learning forecasts can serve as a statistically defensible basis for climate-trigger design. The contribution is not that the thesis fully prices these instruments — Section 5.7 explicitly limits the analysis to a conceptual transferability argument — but that it provides a reproducible forecast-governance layer that can support their design in a more rigorous and regulatory-compliant way.

Chapter 3: Methodology

3.1 Data: E3CL and ERA5

The dataset is E3CL (Extreme Climate Index Library), derived from ERA5 reanalysis and freely accessible from the E3CL Data Station platform. The study area is the Bologna (BO) monitoring station (44.5°N, 11.3°E, elevation 54 m), representative of the Po Valley agricultural region. Six monthly extreme-event indices are analysed over January 1981–April 2025 (n = 532 observations for most variables; n = 488 for forest fire due to December exclusion from fire-season definition). All indices are standardised anomalies with mean 0 and standard deviation 1 over the 1981–2010 baseline. All results in this thesis are computed for this single grid point: the analysis is a point (station-level) forecasting exercise, not a province-wide or spatially aggregated one.

Teleconnection predictors — North Atlantic Oscillation (NAO), East Atlantic/Western Russia pattern (EAWR), Scandinavian pattern (SCAND), and East Pacific/North Pacific (EPNP) — are sourced from the NOAA Climate Prediction Center and appended as lagged exogenous variables at lags 1, 2, 3, 6, 9, 12, 18, and 24 months.

Variable	n	Mean	Std	ACF1	R ² -ceil	Skew	Ex. kurt.	JB p	Signal Quality
Drought	532	−0.049	1.09	0.720	0.903	0.06	−0.05	0.84	Excellent
Forest Fire	489	−0.166	1.06	0.304	0.757	4.03	24.07	<0.001	Good
Extreme Temperature	532	0.367	1.81	0.137	0.608	4.95	38.01	<0.001	Moderate
Hail	532	−0.016	1.16	0.085	0.423	3.39	28.07	<0.001	Poor
Extreme Precipitation	532	0.100	1.45	−0.013	0.063	5.94	52.06	<0.001	Very Poor
Extreme Wind	532	−0.009	0.91	−0.005	0.040	2.76	8.51	<0.001	Very Poor

Table 3.1. Signal diagnostics for six ERA5-derived extreme climate variables at Bologna. ACF1 = lag-1 autocorrelation. R²-ceiling = an illustrative predictability score derived from the autocorrelation of each series after a variance-stabilising transform; a heuristic ranking indicator, not a strict or exactly reproducible bound.

Normality of each standardised index was assessed with the Jarque–Bera (JB) test (Jarque, C. M., & Bera, A. K., 1980), which combines the sample skewness and excess kurtosis into a single statistic that is asymptotically distributed as χ^2 with two degrees of freedom under the null of normality; the last three columns of Table 3.1 report the skewness, excess kurtosis and JB p-value for each variable. The result is clear-cut and has a direct bearing on the pricing step (Sections 3.5–3.6). Drought is the only index consistent with a Gaussian: its skewness (0.06) and excess kurtosis (−0.05) are both negligible, and the JB test does not reject normality (p = 0.84). Every other variable is strongly non-Gaussian, with pronounced positive skewness and heavy tails — extreme precipitation (excess kurtosis 52) and extreme temperature (38) are the most extreme, and even hail, the least persistent series, has an excess kurtosis of 28. For all five non-Gaussian variables the JB statistic runs into the thousands, and normality is rejected at any conventional level (p < 0.001). This is precisely why the closed-form Gaussian exceedance probability of Section 3.6 is applied only to drought; for the remaining variables

the Gaussian Φ is replaced by the empirical conditional exceedance frequency (or a generalised extreme value fit in the tail), which correctly captures their heavier-than-normal probability of large values.

Table 3.1 motivates the modelling choices made in the rest of this chapter, and two of its columns are central. The lag-1 autocorrelation (ACF1) measures how strongly each month's value depends on the previous month; it is a direct proxy for exploitable temporal structure, or memory, in the series. The R^2 -ceiling is a heuristic upper bound on the fraction of variance a linear AR (1) model could explain, and therefore an optimistic indication of how predictable each series is in principle.

This ceiling is a heuristic predictability score derived from the autocorrelation of each series after the diagnostic routine applies a variance-stabilising transform. It always exceeds the square of the raw lag-1 ACF (for drought, 0.903 against a raw ACF1² of 0.52) and is deliberately optimistic; it is used only to rank the six variables by how much exploitable structure they contain, and no quantitative claim rests on its precise value.

Read together, the two columns sort the six hazards into a clear ordering of predictability. Drought sits at the top (ACF1 = 0.720, R^2 -ceiling = 0.903): soil-moisture persistence makes drought months cluster, so even simple models inherit substantial skill. Forest fire and extreme temperature occupy an intermediate band, where a seasonal cycle supplies moderate persistence. At the bottom, hail, extreme precipitation and extreme wind have ACF1 values at or below 0.085 — effectively no month-to-month memory — which means any apparent forecast skill for these series must be treated with suspicion until it is confirmed statistically. This ordering frames the central empirical tension of the thesis: a low ACF1 does not by itself rule out predictability (hail is later shown to be the clearest counter-example), but it does raise the standard of evidence required, and enforcing that standard is precisely the role of the Diebold–Mariano gate introduced in Section 3.4.

3.2 Forecasting Model Library and Selection Framework

Before describing the individual models, it is useful to state the overall forecasting workflow that this chapter implements, because the three subsections that follow correspond to its three stages. The workflow is: (1) a library of sixteen candidate models spanning low- and high-complexity families is assembled for each variable (this section); (2) every model is trained and evaluated with expanding-window walk-forward cross-validation, which repeatedly re-trains the model on all data available up to a point in time and tests it on the next unseen observation (Section 3.3); and (3) each model's out-of-sample forecast errors are compared against a monthly-climatology benchmark using the Diebold–Mariano test, which acts as a statistical gate deciding whether a model's accuracy advantage is real (Section 3.4). Only models that pass this gate are carried forward into the trigger-design and pricing framework. In short: the model library defines what is tried, walk-forward cross-validation defines how it is trained and scored, and the Diebold–Mariano test defines what is believed.

The full pipeline implements 16 model architectures, organised by family into baseline/benchmark, statistical time-series, gradient-boosted tree, hybrid, probabilistic-neural, and ensemble families. For presentation in this thesis, I select the 2 best low-complexity models and 2 best high-complexity models for each variable at each horizon, based on DM test results and RMSE.

Low-complexity models include Monthly Climatology (baseline), Seasonal Naïve, SARIMA (Box et al., 2015), ETS (Hyndman & Khandakar, 2008), RidgeSeasonalDecomp (ridge regression on seasonal decomposition residuals), and Prophet (Taylor & Letham, 2018).

Two of these benchmarks deserve a brief explanation because the entire evaluation is defined relative to them. The Monthly Climatology benchmark is the forecasting equivalent of assuming that next January will look like a typical January: for each calendar month it forecasts the historical mean of that month, computed over the training window (so the forecast for any May is the average of all past Mays observed up to time t). Because the indices are standardised anomalies with a long-run mean of zero, this benchmark is close to — but not identical to — a forecast of zero, and it still captures any residual seasonal mean structure. Crucially, it uses no information about the current state of the climate system, which is exactly what makes it the appropriate 'do-nothing' reference: any model that cannot beat it is adding no value over simply knowing the season. The Seasonal Naïve model is a slightly stronger persistence benchmark: it forecasts each month using the most recent observation from the same month one year earlier (the value at $t-12$). Unlike climatology, it reacts to the latest available same-season value, and it is included to check whether a model's apparent skill comes from genuine learning or merely from exploiting year-on-year seasonal repetition.

Prophet is classified as a low-complexity model despite being a relatively recent forecasting method. The distinction used in this thesis is not based on the age of the method, but on model structure. Like SARIMA and ETS, Prophet is a structured additive time-series model that decomposes a series into trend, seasonality, and calendar effects using a limited number of interpretable parameters. Unlike machine-learning models, it does not learn complex nonlinear interactions from high-dimensional feature spaces. Therefore, Prophet is grouped with the low-complexity models, whereas gradient-boosting and neural-network architectures are classified as high-complexity models.

High-complexity models include: XGBoost (Chen & Guestrin, 2016) in four variants — standard XGBoost; XGBoost_STL; XGBoost_STL, which is itself a mini-ensemble: after an STL decomposition it trains two specialised sub-models — one on the trend and teleconnection features, one on the residual and seasonal-lag features — and averages their forecasts (a 50/50 blend), so each sub-model can concentrate on a different part of the signal; BaggedXGBoost, a variance-reduction ensemble of five independent sub-models trained on bootstrap samples; XGBoost_MV_Spatial, which extends the base model with contemporaneous data from neighbouring stations; and XGBoost_Telecon, which adds NOAA teleconnection indices (NAO, EAWR, SCAND, EPNP) as lagged features. The remaining high-complexity models are LightGBM (Ke et al., 2017), CatBoost (Prokhorenkova et al., 2018), PyTorchLSTM (Hochreiter & Schmidhuber, 1997), and the Ensemble (Bates & Granger, 1969) — an inverse-RMSE-weighted meta-learner combining all constituent forecasts, with weights updated at each walk-forward fold.

Beyond the models highlighted here, the pipeline also implements ARIMA and an ARIMA–XGBoost hybrid (Zhang, 2003), both of which are excluded from the final comparison because of convergence failures.

STL — Seasonal-Trend decomposition using Loess (Cleveland et al., 1990) — referenced by the XGBoost_STL variant splits a time series into three additive components: a smooth trend, a repeating seasonal pattern, and an irregular remainder. XGBoost_STL exploits this by first removing the deterministic trend and seasonal components, which a tree-based learner captures only awkwardly, and then applying gradient boosting to the remainder alone. This lets the machine-learning model

concentrate its capacity on the genuinely hard-to-predict residual variation rather than re-learning the obvious annual cycle, which in practice improves stability at the seasonal horizons.

To be explicit about geographic scope: every model in this thesis forecasts the index for the single Bologna (BO) grid point only. This is a point (station-level) forecasting problem, not a province-wide or regionally pooled one. The one apparent exception is XGBoost_MV_Spatial, which adds the two or three most strongly correlated neighbouring-station series for the same variable as extra predictors; even there, however, the quantity being forecast is still the Bologna index alone — the neighbouring stations enter only as explanatory inputs, not as additional forecast targets. Spatial generalisation to other Emilia-Romagna stations is left to future work (Section 6.3).

Empirical results in Chapter 4 show that the Ensemble and STL-enhanced variants consistently rank among the best performers, while PyTorchLSTM exhibits high variance — superior at short horizons for high-signal series but catastrophically poor for near-white-noise series at long horizons.

The selection principle is to present the best-performing low-complexity model and the best-performing high-complexity model for each variable-horizon combination, evaluated using the DM-test and RMSE criteria described above. For variables where the DM test confirms skill, the best models are those with DM-positive statistics. For unpredictable variables, the best models are those that fail least severely.

3.3 Walk-Forward Cross-Validation

This is the second stage of the workflow set out at the start of Section 3.2: having defined the model library, we now specify exactly how each model is trained and scored out of sample.

All models are evaluated using expanding-window walk-forward cross-validation, starting from a minimum training window of 150–250 observations (150 for $h = 1$ and $h = 3$, rising to 200 for $h = 6$ and 250 for $h = 12$, so that the longer horizons are trained on more history). At each fold, the model is trained on all available data up to date t , and the h -step-ahead forecast is evaluated against the held-out observation at $t+h$. The maximum number of folds is capped at 50 for computational tractability; when the full walk-forward path would exceed 50 folds, they are sub-sampled evenly across the record. A purge gap prevents temporal data leakage in the hyperparameter tuning loops: the standard evaluation folds leave no gap, while the inner Optuna tuning loop enforces a purge of $\max(h, 6)$ observations plus a two-step embargo between training and validation, following the purged cross-validation scheme of López de Prado (2018).

The 12 months from May 2025–April 2026 serve as a genuine out-of-sample hold-out period, used only in the forecast-accuracy visualisations. For each horizon, the walk-forward procedure generates up to 50 evaluation folds, but these folds overlap in calendar time: at $h = 12$, for example, the forecast for $t+12$ issued at time t and the forecast for $t+12$ issued at time $t-1$ both contribute errors for the same target month. When computing DM statistics, the effective number of non-overlapping test observations is therefore smaller than the raw fold count. Table 3.2 reports, for each horizon h , the approximate number of folds generated and the corresponding number of non-overlapping observations available for DM inference.

Horizon h	Min. Training Window	Max. Folds	Non-overlapping Obs.
1 month	250 obs.	50	≈ 50 (no overlap)
3 months	250 obs.	50	≈ 17 (overlap = 2)
6 months	200 obs.	50	≈ 9 (overlap = 5)
12 months	150 obs.	50	≈ 5 (overlap = 11)

Table 3.2. Walk-forward design and approximate effective sample sizes by horizon. Non-overlapping observations are computed as $\max(1, \text{floor}(50/h))$. Exact counts should be verified from the CV implementation output; these figures illustrate the order of magnitude of the effective sample available for DM inference.

3.4 The Diebold-Mariano Test: A Statistical Criterion for Forecast Evaluation

The Diebold–Mariano (DM) test (Diebold & Mariano, 1995) is the primary forecast-evaluation criterion used in this thesis. Its job is narrow but decisive: to decide whether a candidate model forecasts the index more accurately than the climatological benchmark by a margin that is statistically significant within the evaluated sample, rather than by chance.

The intuition is a paired comparison of errors. At every walk-forward fold the candidate model and the benchmark each produce a forecast for the same target month, and each incurs a forecast error. Squaring those errors and taking their difference gives, for each fold, a loss differential — a single number that is positive when the benchmark did worse than the candidate and negative when it did better. The DM test asks whether the average loss differential is significantly different from zero, using a variance estimate that corrects for the autocorrelation introduced by the overlapping forecasts described in Section 3.3. A positive DM statistic therefore means the model generates lower average squared errors than the naive benchmark; the associated p-value quantifies how confident we can be that this advantage is not an artefact of sampling.

This division of labour between RMSE and the DM test is central to the thesis. RMSE quantifies the magnitude of improvement, but the DM test determines whether that improvement is statistically meaningful. An 8% gain over climatology may simply reflect noise or overfitting, and only the DM test can confirm its robustness. In this thesis, it acts as a statistical gate, ensuring that only models with genuinely significant gains are used to define trigger thresholds and feed the pricing calculation. A model can have a lower RMSE than climatology and still fail the gate — and when it does, that lower RMSE is treated as unproven and is not used.

Formally, the test compares the squared-error loss of the candidate against that of Monthly Climatology. The loss differential at each evaluation point is the climatology's squared error minus the model's squared error, and the statistic divides the mean of this differential by a heteroskedasticity- and autocorrelation-consistent (HAC, Newey–West) (Newey, W. K., & West, K. D, 1987) standard error computed over lags 1 to $h-1$ — the correction appropriate for h -step-ahead forecasts, whose errors are serially correlated. The resulting statistic is compared against a Student t-distribution with $n-1$ degrees of freedom, two-sided.

The primary significance threshold is $\alpha = 0.05$ (two-sided), applied directly: a model is judged to beat climatology when its DM statistic is positive, and its p-value falls below 0.05. Stronger evidence is flagged at the conventional levels $\alpha = 0.01$ and $\alpha = 0.001$, so that a result at $p < 0.001$ (for example the hail Ensemble at $h = 12$) rests on much firmer ground than one at $p < 0.05$ (for example the drought

Prophet result at $p = 0.044$). The shorthand 'DM-confirmed' is used in the results chapter for any model significantly better than climatology at $\alpha = 0.05$, and 'significantly worse' is used symmetrically for negative results.

On this standard, the pipeline yields 9 statistically confirmed positive results across the 24 variable–horizon combinations (six variables $\times$ four horizons) — roughly 37% — while for the remaining combinations the climatological mean stays the best available forecast. Two cautions accompany the test. First, with only about 100 non-overlapping hold-out observations per horizon (Table 3.2), it has limited power: a sizeable RMSE gain can still fall short of significance, so a non-significant result means 'not demonstrated' rather than 'no skill.' Second, when a model produces near-constant predictions across some folds the Newey–West denominator can collapse and the DM statistic diverges to implausible magnitudes (an absolute value above 1000); such cases are flagged as numerical artefacts and excluded from interpretation rather than read as extraordinary skill.

Because several models are screened within each variable–horizon cell, the confirmed results are best read as flags for closer inspection rather than as guarantees. Where family-wise error is a concern, a Bonferroni-style correction can be reported as a stricter robustness check; the primary evidentiary standard used throughout the results, however, is the unadjusted $\alpha = 0.05$ described here, which is the threshold underlying the count of confirmed results.

3.5 Trigger Design and Conditional Exceedance Probability

The contractual trigger in this thesis is a fixed climatological return-period level, $\tau = P_q$, defined as the q -th percentile of the historical index distribution. For drought, $q = 0.90$ gives $\tau \approx +1.37$ (a one-in-ten level), and a more catastrophic design would use $q = 0.95$ ($\tau \approx +1.74$).

It is worth being explicit about how a return period is obtained, since it is what fixes the numerical trigger. A return period is simply the average waiting time between exceedances. If the trigger sits at the q -th percentile of the index, the probability that the index exceeds it in any single evaluation period is $1 - q$, and the return period T is the reciprocal of that probability:

$$\tau = P_q \quad , \quad Pr(I > \tau) = 1 - q$$

$$T = \frac{1}{1 - q}$$

So, $q = 0.90$ corresponds to an exceedance probability of 0.10 and a one-in-ten return period, and $q = 0.95$ to 0.05 and a one-in-twenty return period. Because the index is standardised to mean 0 and standard deviation 1, these percentiles also map onto z-scores: under exact normality $P90 \approx +1.28$ and $P95 \approx +1.64$. The slightly higher empirical values used here (+1.37 and +1.74) are read directly from the realised drought distribution rather than from the normal curve, and their small excess over the normal z-scores reflects the mild right-skew of the index. One point of wording: because the index is analysed at monthly resolution, the P90 level is by construction exceeded in about 10% of months (a roughly ten-month return period); the “one-in-ten-year” reading holds only if the trigger is assessed once per relevant decision period (for example the seasonal index value the contract references). The contract should state the evaluation window to which the return period applies, so the two descriptions stay consistent.

Defining the trigger by return period makes it transparent and auditable and answers directly why a particular numerical value is chosen. Forecast information does not move the trigger; it updates the conditional probability that the realised index will cross it.

Let $\hat{\mu}$ be the DM-validated point forecast at horizon h (written $\hat{\mu}_{t+h}$) and σ_f the forecast-error standard deviation at that horizon, estimated from the walk-forward forecast errors. Under the assumption of approximately unbiased forecast errors, the RMSE reported in Appendix B is used as a practical estimate of σ_f ; if the forecast is materially biased, $\text{RMSE}^2 = \text{variance} + \text{bias}^2$ overstates the error variance, so the error standard deviation (or a bias-corrected forecast) should be used. The conditional crossing probability is:

$$Pr(I > \tau \mid \hat{\mu}_{t+h}, \sigma_f) = 1 - \Phi\left[\frac{\tau - \hat{\mu}_{t+h}}{\sigma_f}\right]$$

Here Φ denotes the cumulative distribution function (CDF) of the standard normal distribution, $\Phi(z) = Pr(Z \leq z)$ for $Z \sim N(0,1)$; the expression $1 - \Phi[(\tau - \hat{\mu})/\sigma_f]$ is therefore the probability that a Gaussian forecast with mean $\hat{\mu}$ and standard deviation σ_f lands above the trigger τ . The Gaussian form is not assumed universally. It is used only where the standardised index is statistically consistent with normality, which is checked per variable with the Jarque–Bera test (Section 3.1). Drought is the one series that passes (JB $p = 0.84$, so normality is not rejected at any conventional level), which is why the analytic Gaussian expression is applied to it; for the non-Gaussian variables the Gaussian Φ is replaced by the empirical conditional exceedance frequency, as described in Section 3.6.

The forecast prediction band describes forecast uncertainty and feeds both this pricing probability and the anticipatory-warning logic of the hybrid product (Section 6.4):

$$\hat{\mu}_{t+h} \pm k \cdot \sigma_f$$

Three standard-deviation-like quantities are kept distinct: σ_{hist} (unconditional index standard deviation, ≈ 1.088 for drought, used only in the static baseline); σ_f (forecast-error standard deviation, the conditional spread); and Δ (cross-model disagreement, ≈ 0.085 for drought, a diagnostic and optional premium loading). Δ is **not** used to set the trigger.

Once a model has been DM-confirmed, its forecasts are used to estimate the conditional probability of crossing the fixed insurance trigger, exactly as set out above. A moving-threshold variant — in which the forecast is allowed to shift the trigger itself, rather than only the crossing probability — is a conceptual extension that requires granular loss data to calibrate. To keep this chapter focused on the methods implemented and priced, that extension is not developed here; it is presented as future work in Section 6.4.1.

3.6 Insurance Design: Premium Under Dynamic Triggers

Given the fixed trigger τ , the expected payout is

$$E[L] = P \cdot Pr(I > \tau \mid F_t)$$

where P is the contractual payout — the fixed sum insured that is paid when the trigger is breached (the €1 million notional used in the worked examples of Chapter 5). $E[L]$ is therefore the expected payout: the payout amount multiplied by the probability that the trigger is crossed.

The static baseline uses the unconditional distribution, $\Pr(I > \tau \mid \mu_{hist}, \sigma_{hist})$, which by construction equals the tail probability of the chosen percentile (10% at P90). The forecast-informed price uses the conditional distribution centred on the forecast,

$$\Pr(I > \tau \mid \hat{\mu}_{t+h}, \sigma_f) = 1 - \Phi\left[\frac{\tau - \hat{\mu}_{t+h}}{\sigma_f}\right]$$

Because the forecast mean enters the numerator, an elevated forecast raises the crossing probability and hence the premium, and a benign forecast lowers them.

The premium is

$$P_{premium} = (1 + \lambda) \cdot E[L]$$

with λ the loading factor (0.20–0.40 in the examples). For drought the probability is computed analytically, because the drought index is statistically indistinguishable from a Gaussian: the Jarque–Bera (JB) test — a normality test built from the sample skewness and kurtosis, asymptotically distributed as χ^2 with two degrees of freedom under the null of normality — returns $p = 0.84$, so normality cannot be rejected. For the non-Gaussian variables, the Gaussian Φ is replaced by the empirical conditional exceedance frequency, which is heavier in the tail.

Normality was screened for every variable with the JB test as part of the signal diagnostics (Section 3.1). Drought is the only series that passes; the other five are markedly non-Gaussian — hail most severely, with an excess kurtosis of roughly 28, an extremely heavy right tail for which a Gaussian would badly understate the exceedance probability. For those variables the conditional crossing probability is therefore read from the empirical distribution of the walk-forward forecast errors (or, in the tail, a generalised extreme value fit) rather than from Φ ; the relevant cases are flagged where they arise (Sections 4.4–4.7 and 5.4).

An optional additional loading for model disagreement may be applied by scaling λ upward when the cross-model spread Δ_{t+h} is large — reflecting elevated forecast uncertainty in that period — but this is a discretionary underwriting adjustment and does not alter the exceedance-probability calculation itself.

3.7 Methodological Limitations and Statistical Caveats

This section documents the key statistical limitations of the forecasting pipeline that qualify the interpretation of results in Chapter 4. Five limitations bear directly on the interpretation of results.

First, Although the full dataset spans 532 monthly observations (January 1981–April 2025), the effective sample size available for forecast evaluation declines substantially at long horizons. At $h=12$, the walk-forward cross-validation procedure is capped at 50 evaluation folds for computational reasons, of which only about 4–5 non-overlapping observations are available for DM inference at $h = 12$ (max (1, floor (50/12))). This is far below the 50-observation setting typically considered in power studies, so DM tests at $h=12$ have very low power, and their results must be treated as exploratory. Gonçalves and Kilian (2004) show that with around 50 evaluation observations, DM test power to detect a 10% RMSE improvement can still fall to roughly 40–60%. Given that this thesis has only about 5 non-overlapping observations at $h=12$, the effective power is substantially lower.

Second, multi-step forecasting generates overlapping prediction errors: the 12-month-ahead forecast issued at time t and the forecast issued at $t+1$ both use observations from the same future period. This induces moving-average serial correlation of order $h-1$ in the sequence of forecast errors, violating the assumption of independent errors that underlies the standard DM test. This thesis addresses this through Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey & West, 1987), with a bandwidth of $\max(1, \text{floor}(1.5 \times h))$ applied in the DM statistic calculation. However, finite-sample properties of HAC-corrected DM tests with overlapping horizons are known to be imperfect, and numerical instabilities (DM statistics exceeding 10,000 in absolute value) arising from Newey–West denominator collapse are excluded from inference and flagged in the results.

Because of HAC bandwidth adjustment, the effective degrees of freedom used in DM p-value computation vary across variable–horizon cells, meaning the DM statistic-to-p-value mapping is not monotonic across horizons. A DM statistic of 3.4 at $h=3$ can yield a larger p-value than a DM statistic of 3.3 at $h=12$ if the former uses fewer effective df due to a wider HAC bandwidth. This is expected behaviour and is documented where relevant in Chapter 4.

Third, All DM test results in this thesis are computed relative to the Monthly Climatology benchmark, defined as the recursive historical mean for each calendar month. This choice is standard in seasonal climate forecasting literature (Wilks, 2019) and provides a transparent, reproducible baseline. However, results are conditional on this specific benchmark. Alternative benchmarks — such as seasonal naïve (last year’s same-month value), persistence (most recent observation), or a linear AR(1) model — may produce materially different relative performance rankings for some variable-horizon combinations. In particular, the AR (1) benchmark would impose a higher performance bar for high-autocorrelation series such as drought ($\text{ACF1} = 0.720$) and might reduce the apparent DM advantage of complex models at short horizons. The choice of Monthly Climatology as the sole benchmark is therefore a deliberate simplification rather than a claim that it is the most demanding standard. For high-autocorrelation series such as drought, an AR (1) or low-order ARIMA benchmark would capture a large share of the available short-horizon predictability, and DM statistics relative to such baselines would likely be smaller in magnitude. In this thesis, all reported DM results should thus be read as conditional on the climatology benchmark: they demonstrate that certain ML models beat a naïve seasonal mean, but they do not yet prove superiority over more structured time-series baselines. A full robustness exercise against AR (1) and seasonal-naïve benchmarks, starting with the flagship drought result, is identified as a priority for future work rather than presented as already solved here.

Fourth, the multiple-testing burden is substantial and must temper how the nine nominal results are read. With 24 hazard–horizon cells and up to 16 models per cell, roughly 384 hypothesis tests are performed; under a strict global null about 19 would reach the 5% level by chance, of which roughly half — nine or ten — would fall in the positive tail. The nine positive results are therefore close to the number chance alone would generate, so the raw count is not in itself evidence of skill. What the multiplicity argument cannot explain is the strength of the two leading results: drought (Ensemble, $h = 12$, $p = 0.002$) and hail (Ensemble, $h = 12$, $p = 0.0002$) both survive a conservative Bonferroni correction ($\alpha^* = 0.05/16 \approx 0.0031$) and are far too significant to be artefacts of testing 384 combinations. These two are treated as the robust findings of the thesis; the remaining seven are reported as suggestive evidence that motivates further testing rather than confirmed skill.

Fifth and finally, all forecasts at horizons $h \in \{1, 3, 6, 12\}$ are constructed as direct multi-step forecasts: a separate model is trained and evaluated for each target horizon h , and predictions are made directly h steps ahead rather than by recursive one-step-ahead chaining. This approach avoids compounding of one-step errors, which can be severe for nonlinear models at long horizons, but produces h separate model instances per variable ($4 \times 16 = 64$ model-horizon combinations per variable). The Ensemble meta-learner is constructed from the constituent point forecasts at each specific horizon h ; Ensemble weights are updated at each walk-forward fold using inverse-RMSE weighting computed on out-of-sample constituent forecasts, ensuring the meta-learner itself is trained on held-out predictions. Models are trained against mean squared error (MSE/RMSE) as the primary loss function, apart from Quantile XGBoost (which optimises pinball/quantile loss at the 10th, 50th, and 90th percentiles) and DeepAR (which maximises log-likelihood of a parametric predictive distribution). (Salinas et al., 2020)

Chapter 4: Empirical Results

This chapter presents results for all six climate variables. For each variable, I show: (i) signal quality and why it matters for forecast ability; (ii) the 2 best low-complexity and 2 best high-complexity models identified by DM test and RMSE; (iii) supporting figures; and (iv) the DM-confirmed result that feeds into Chapter 5's insurance design.

The presentation strategy in this chapter is deliberately layered. For each variable, summary diagnostics establish whether the series is structurally forecastable; model-comparison tables and heatmaps then show which architectures perform best across horizons; walk-forward and hold-out figures are included not as standalone illustrations, but as diagnostic evidence on forecast stability, horizon sensitivity, and operational plausibility. This sequence is important because DM significance alone does not fully describe usability for insurance design: a model may outperform climatology statistically while still being economically fragile, horizon-specific, or dependent on non-temporal information such as spatial interpolation.

4.1 DM Results Summary

Across 24 variable-horizon combinations, 9 model-horizon results return statistically significant positive DM results (spanning 8 distinct cells; Drought at $h=12$ has two confirmed models), Table 4.1 lists these nine confirmed results, with the two Bonferroni-robust cases (drought and hail at $h = 12$) highlighted:

Variable	Horizon	Best Model	DM Stat	p-value
Drought	$h = 12$	Ensemble	3.252	0.002 **
Drought	$h = 12$	Prophet	2.068	0.044 *
Forest Fire	$h = 3$	XGBoost_STL	3.423	0.019 *
Extreme Temperature	$h = 6$	SARIMA	2.211	0.032 *
Extreme Wind	$h = 1$	XGBoost_MV_Spatial	2.088	0.042 *
Extreme Wind	$h = 3$	BaggedXGB	2.506	0.016 *
Hail	$h = 3$	Ensemble	2.783	0.013 *
Hail	$h = 6$	Prophet	2.607	0.012 *
Hail	$h = 12$	Ensemble	4.743	0.0002 ***

*Table 4.1. All 9 DM-confirmed positive results across 24 variable-horizon combinations. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. vs. Monthly Climatology benchmark. Only drought and hail at $h=12$ survive Bonferroni family-wise correction ($\alpha = 0.0031$, see section 3.4); all other results constitute tier-(ii) suggestive evidence.*

A consistent pattern across variables is the rapid decay of skill with lead time. The strongest DM statistics for drought and hail occur at $h = 12$, but only in a very small effective evaluation sample and with the benchmark-dependence and multiple-testing caveats discussed in Section 3.7. The preference for $h = 1$ and $h = 6$ horizons in insurance design is established in 1.6.

4.2 Drought

Signal quality: ACF1 = 0.720, R^2 -ceiling = 0.903. Excellent — the strongest predictable series in the portfolio. Multi-month soil moisture persistence means drought months tend to cluster: a drought month is highly likely to be followed by another drought month. This strong autocorrelation means even simple models can perform well, and the high ceiling gives complex models room to improve.

Best low-complexity models: SARIMA ($h=1$ RMSE=0.918, $R^2=0.267$) and ETS ($h=1$ RMSE=1.031, $R^2=0.076$). SARIMA exploits the seasonal autocorrelation structure cleanly.

Best high-complexity models: PyTorchLSTM ($h=1$ RMSE=0.599, $R^2=0.313$) and Ensemble ($h=1$ RMSE=0.607, $R^2=0.294$). At $h=12$, the Ensemble becomes the DM-confirmed champion with RMSE=1.131 vs climatology 1.225.

Key DM result: Ensemble ($h=12$): DM=3.252, $p=0.002$. Prophet ($h=12$): DM=2.068, $p=0.044$. These are the only two DM-confirmed models for drought, and only at the 12-month horizon. At shorter horizons ($h = 1, 3, 6$) the RMSE improvements are even larger — the Ensemble achieves RMSE = 0.607 at $h = 1$ versus climatology 1.123, a 46% reduction — yet DM significance is not reached. This is not a small-sample effect ($h = 1$ in fact has the most evaluation folds); DM significance depends on the mean loss differential relative to its standard error, not on the size of the RMSE gain. At $h = 1$ the per-period advantage is large but highly variable and serially correlated, so the Newey–West (HAC) correction inflates its standard error and lowers the effective degrees of freedom, leaving DM = 1.12 ($p = 0.28$) despite the large average gain. At $h = 12$ the advantage is smaller in RMSE terms but far more stable relative to its standard error, which is why it — and not the short-horizon gains — clears the gate.

Important failure: ETS is confirmed significantly WORSE than climatology at $h=3, 6, 12$ (DM values -2.32 to -4.44). ETS over-projects recent trends, actively harming performance at seasonal leads. RidgeSeasonalDecomp is also confirmed worse at $h=3$ (DM= -3.33). These models should not be used in drought trigger design based on current evidence.

Figure 4.1 maps RMSE for all sixteen models across the four horizons as a heatmap, making the cross-model pattern visible at a glance: the Ensemble and PyTorchLSTM occupy the lowest-error cells at $h=1$, the Ensemble retains the lowest-error column at $h=12$, and the trend-extrapolating models (ETS, RidgeSeasonalDecomp) migrate to the high-error end of the scale as the horizon lengthens. The heatmap therefore corroborates visually the headline finding that no single model dominates at every horizon, yet the Ensemble is never far from the frontier.

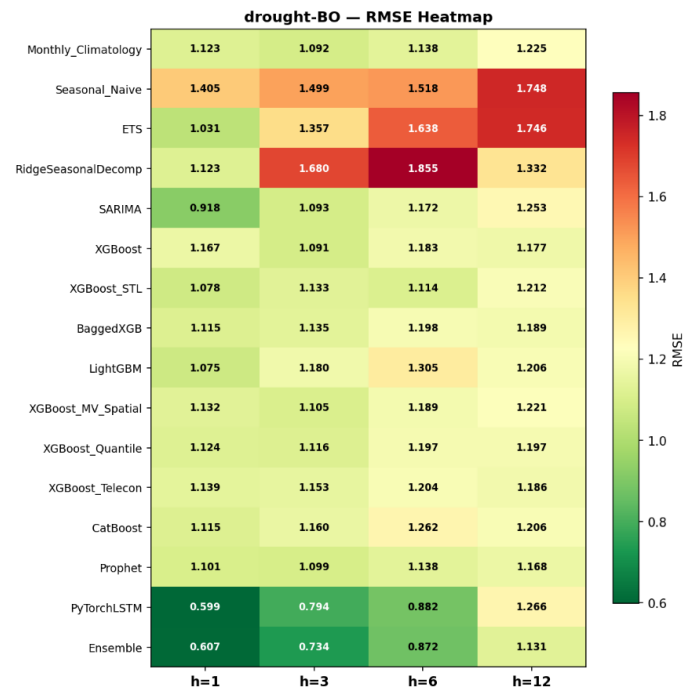


Figure 4.1. RMSE heatmap for drought across all 16 models and 4 horizons.

Figure 4.2 traces the $h = 12$ walk-forward forecast path against the realised drought index over the full 1981–2025 evaluation window, with the Ensemble (RMSE = 1.131) and the Monthly Climatology benchmark (RMSE = 1.225) overlaid. The Ensemble line follows the broad multi-year swings of the index more closely than the near-flat climatology line, which is the visual counterpart of its DM-confirmed advantage; the gap is modest but systematic, consistent with the damped-amplitude behaviour expected at annual leads.

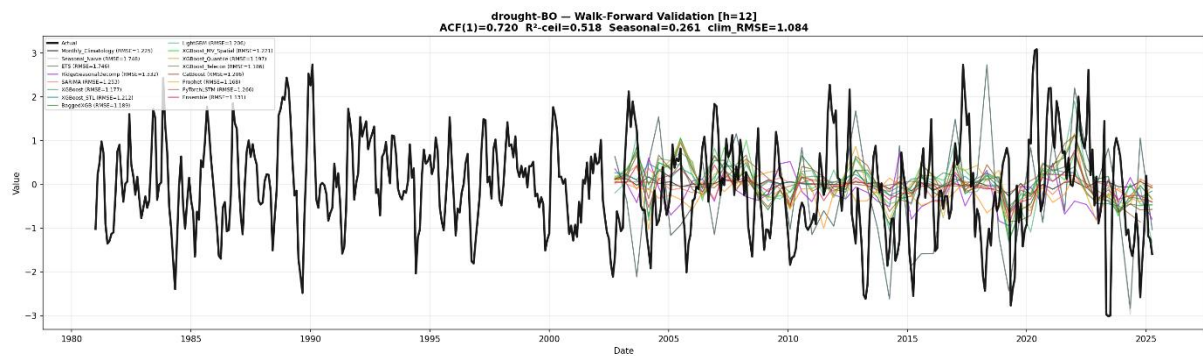


Figure 4.2. Walk-forward validation for drought at $h=12$ (1981–2025).

Figure 4.3 isolates the genuine out-of-sample period (May 2025–April 2026), showing the Ensemble and Prophet hold-out forecasts that are carried into the Chapter 5 pricing example. Both achieve the lowest hold-out RMSE, confirming that the in-sample advantage of these two models survives into data the model never saw during training — the key requirement before either can be used operationally.

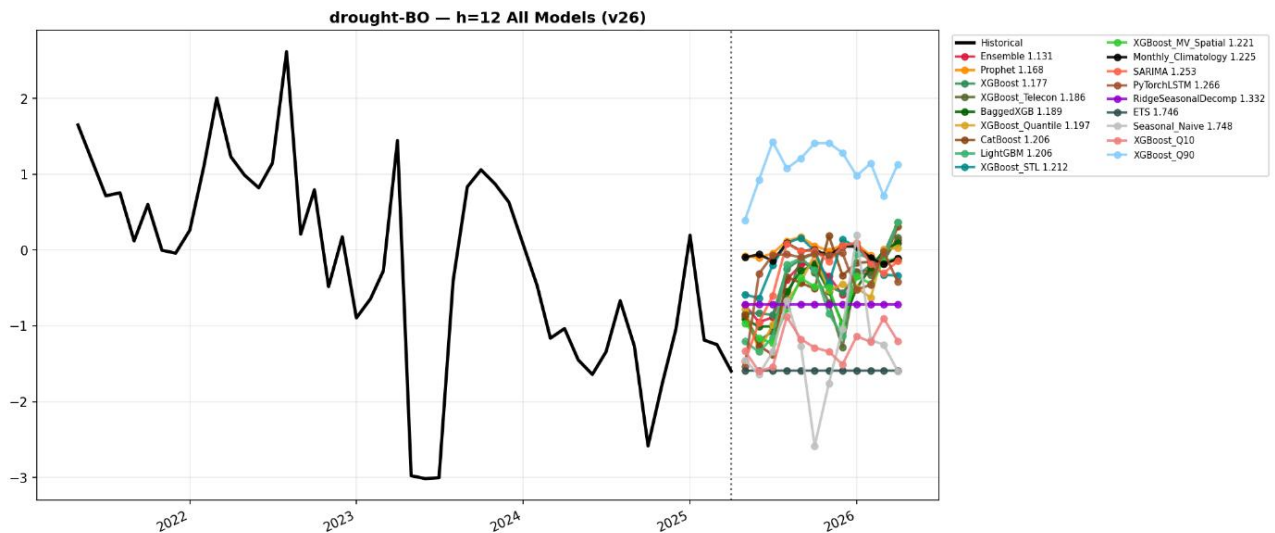


Figure 4.3. Hold-out forecasts for drought at $h=12$ (May 2025–April 2026). Ensemble and Prophet achieve lowest RMSE. These forecasts are used directly in the Chapter 5 insurance example.

Figure 4.4 summarises how the three-error metrics evolve with lead time. The Ensemble holds the lowest RMSE curve across all horizons, whereas the ETS and RidgeSeasonalDecomp curves deteriorate sharply from $h = 3$ onward — the same divergence that appears above as the confirmed “significantly worse” results. Reading the three panels together shows that the Ensemble's edge is consistent across RMSE, MAE and R^2 , rather than being an artefact of any single metric.

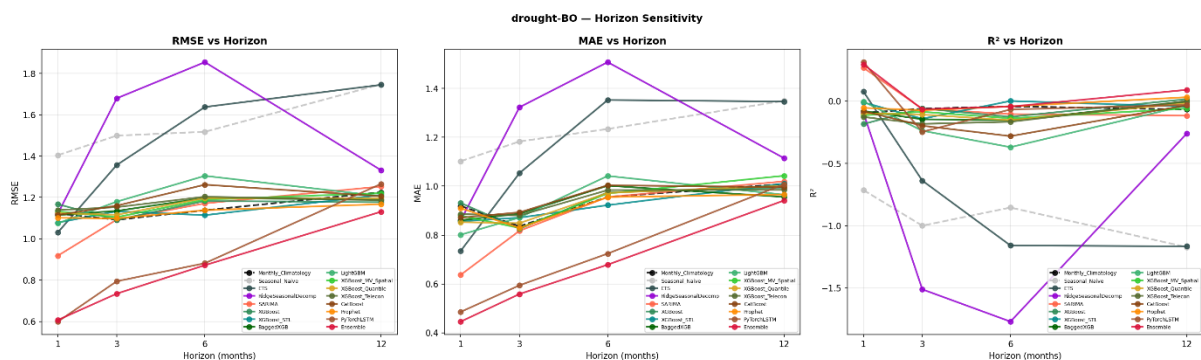


Figure 4.4. RMSE, MAE, and R^2 vs forecast horizon for drought. Ensemble maintains lowest RMSE across all horizons.

Figure 4.5 plots the lagged correlations between the drought index and the teleconnection indices, providing the empirical justification for including those predictors in the XGBoost and Ensemble models: EAWR is positively correlated at lags of 2–3 months (+0.29 to +0.34) and SCAND negatively correlated at lags of 1–3 months (−0.32 to −0.14). These are the strongest and most coherent teleconnection signals of any variable in the study, which helps explain why drought is also the most forecastable series.

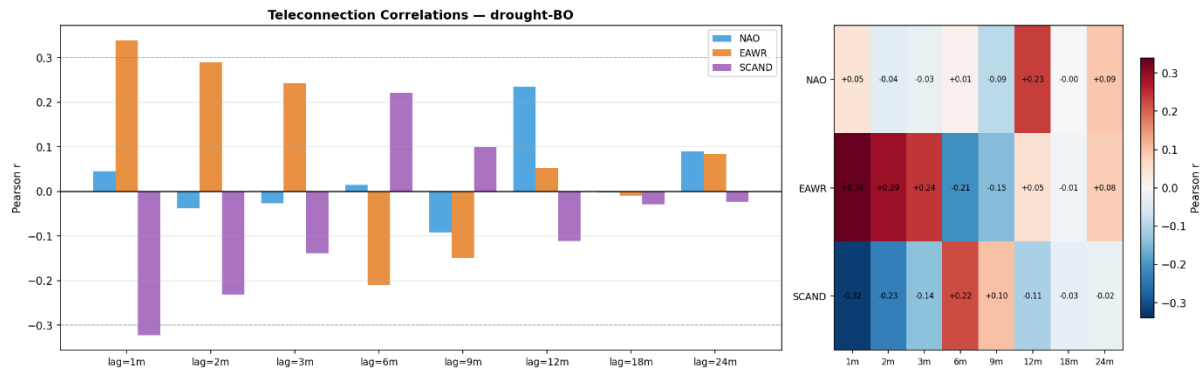


Figure 4.5. Teleconnection correlations for drought.

4.3 Hail

Signal quality: ACF1 = 0.085, R^2 -ceiling = 0.423. Poor — hail is dominated by convective processes with low month-to-month memory. Yet it produces the largest number of DM-confirmed results in the entire study (3 positive confirmations at $h=3, 6, 12$). This apparent paradox is the most statistically notable finding in this study.

Best low-complexity models: ETS (approaching significance at $h=3$, DM=1.91, $p=0.06$) and SARIMA (consistent across horizons). Neither achieves DM confirmation, but ETS avoids the catastrophic failures seen in drought.

Best high-complexity models: Ensemble (DM-confirmed at $h=3$ and $h=12$) and Prophet (DM-confirmed at $h=6$). These are the only architectures that capture the latent seasonal predictability of hail.

Key DM results: Ensemble ($h=12$): DM=4.743, $p=0.0002$ *** — the strongest positive DM statistic in the entire study. Prophet ($h=6$): DM=2.607, $p=0.012$ (tier-ii, suggestive). Ensemble ($h=3$): DM=2.783, $p=0.013$ (tier-ii, suggestive).

Critical failure: PyTorchLSTM hail records its catastrophic confirmed failure at $h = 6$ (DM = -8.121; RMSE 1.483 vs climatology 1.141, +30% worse) and is non-significant relative to climatology at $h = 12$ (RMSE 1.003, marginally better).

Mechanism: The source of the Ensemble's annual hail skill is not confirmed by linear diagnostics — all EAWR-hail correlations are non-significant ($p>0.24$). The mechanism is unknown and requires SHAP feature-importance analysis to resolve. The DM-confirmed result is robust; the physical explanation is pending. (Lundberg & Lee, 2017)

Figure 4.6 presents the hail RMSE heatmap, and it places the best and the worst behaviour in the chapter side by side for the same low-signal variable. The Ensemble column is lowest-error at the long horizons (RMSE = 0.898 at $h = 12$ against climatology's 1.072), while PyTorchLSTM produces the single worst cell in the chapter at $h = 6$ (RMSE 1.483, +30% worse than climatology) — the visual signature of its confirmed catastrophic failure.

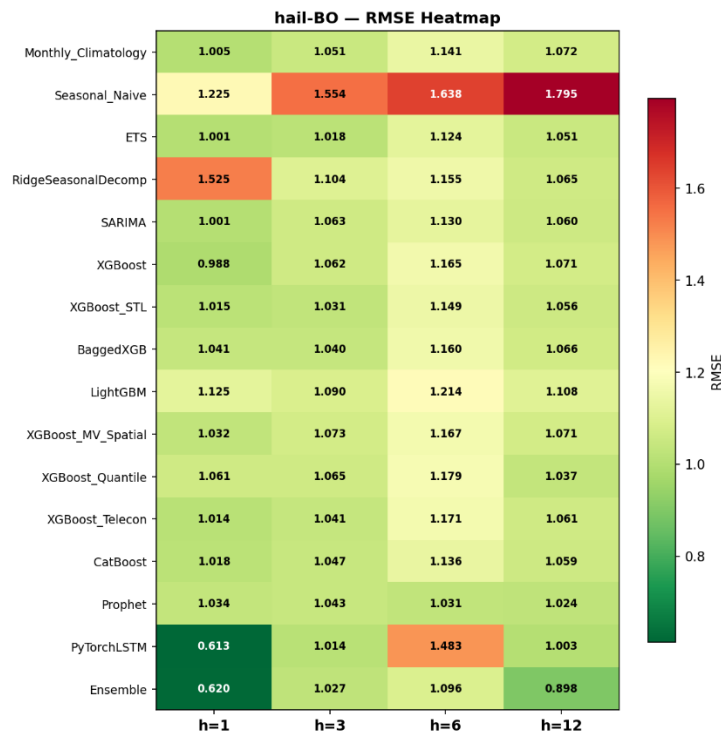


Figure 4.6. RMSE heatmap for hail.

Figure 4.7 shows the $h = 12$ walk-forward path for hail, with the Ensemble (RMSE = 0.898) running clearly below the climatology line (1.072). Given hail's near-zero month-to-month autocorrelation, this visible tracking is the empirical basis for the “predictability paradox” noted above and the single most surprising result in the chapter.

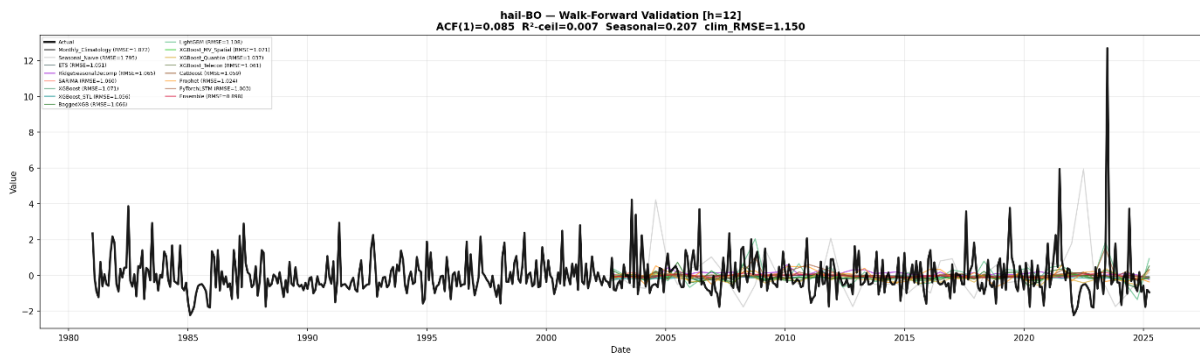


Figure 4.7. Walk-forward validation for hail at $h=12$.

Figure 4.8 plots horizon sensitivity. The Ensemble maintains its skill across $h = 3, 6$ and 12 , whereas PyTorchLSTM's RMSE spikes at $h = 6$ (though not DM-confirmed there) before its confirmed failure at $h = 12$. The figure makes clear that the Ensemble's advantage is horizon-robust, in contrast to the brittle, horizon-specific behaviour of the deep-learning model.

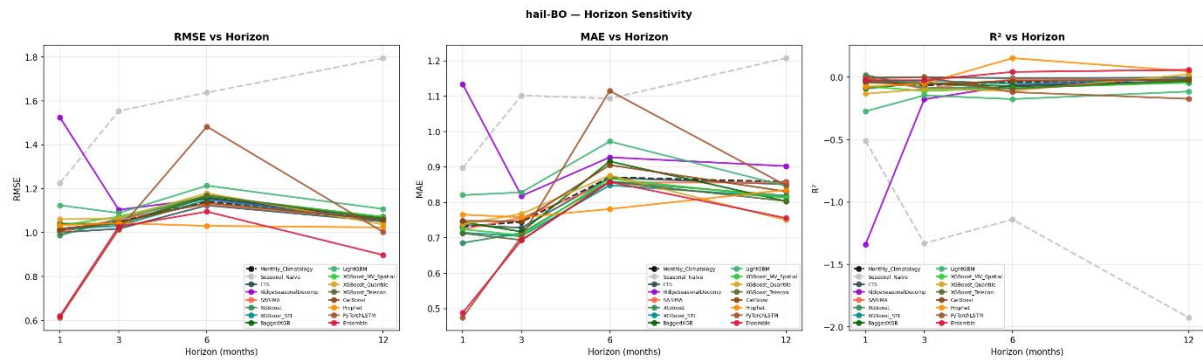


Figure 4.8. Horizon sensitivity for hail.

Figure 4.9 plots the EAWR–hail lagged correlations and, in deliberate contrast to the drought case (Figure 4.5), every correlation here is small (+0.08 to +0.14) and statistically non-significant. The figure is included precisely to document the absence of a linear teleconnection mechanism, reinforcing the point that the Ensemble's confirmed annual hail skill is not explained by the linear diagnostics and remains an open question pending SHAP analysis.

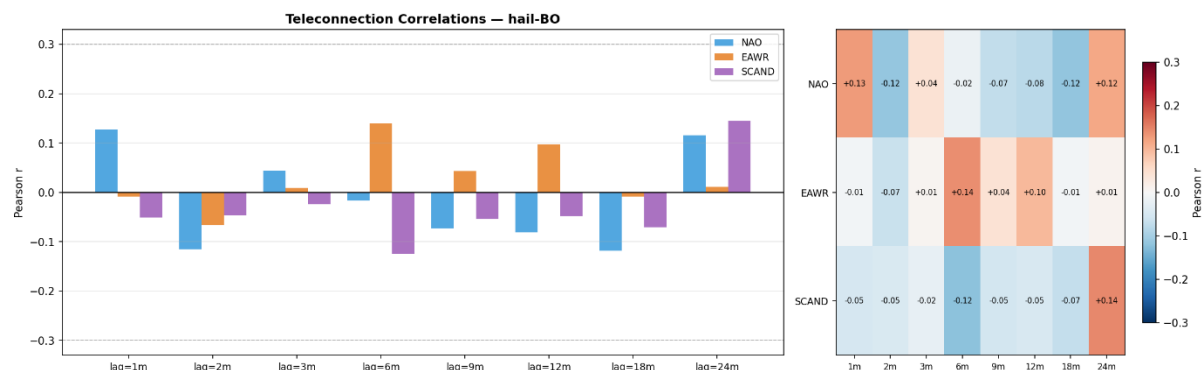


Figure 4.9. Teleconnection correlations for hail.

4.4 Forest Fire

Signal quality: ACF1 = 0.304, R^2 -ceiling = 0.757. Good — seasonal fire weather cycle provides moderate persistence.

Best low complexity: SARIMA ($h=1$ RMSE=0.607) and Prophet ($h=1$ RMSE=0.587). Both competitive but non-significant at $h=1$.

Best high complexity: XGBoost_STL (DM-confirmed at $h=3$: DM=3.423, $p=0.019$) and Ensemble (best RMSE at $h=1$: 0.305).

Key DM result: XGBoost_STL ($h=3$): DM=3.423, $p=0.019$ — the only DM-confirmed (tier-ii, suggestive) positive result for forest fire. Its STL decomposition separates the deterministic seasonal fire-season cycle from the residual variation, allowing targeted skill at the 3-month lead.

Note on the DM reference distribution: significance is assessed against a t-distribution with $n - 1$ degrees of freedom, where n is the number of valid (gap-free) forecast pairs available at that horizon, combined with a HAC/Newey–West variance and the small-sample correction of Harvey, Leybourne and Newbold (1997). The number of valid pairs varies sharply across cells: drought at $h = 12$ retains

the full set of folds ($n \approx 50$, $df \approx 49$), giving the Ensemble result $p = 0.002$, whereas the forest-fire record excludes December and is therefore fragmented, leaving few valid pairs at $h = 3$ ($n \approx 6$, $df \approx 5$). This is why XGBoost_STL at $h = 3$ ($DM = 3.42$) reaches only $p = 0.019$: under a standard-normal or large-sample reference the same statistic would imply $p \approx 0.001$ and would clear Bonferroni, but under the correct small-sample reference it does not — so the Bonferroni-robust claim is restricted to drought and hail at $h = 12$. The SARIMA $h = 3$ value ($DM \approx 25,900$, $p \approx 0$) reflects a HAC denominator collapse and is excluded from inference.

At $h = 12$, the Newey–West HAC estimator collapses for several models, producing unstable DM statistics with denominators close to zero; these values are therefore excluded from inference. Given the limited number of non-overlapping evaluation observations at this horizon and the numerical instability of the HAC adjustment, this thesis does not attempt to rescue the forest-fire $h = 12$ result using alternative DM implementations. A bootstrap-based DM test, as noted in Section 5.5.1, would be a natural next step to explore the robustness of long-horizon skill, but was beyond the scope of the current project.

Additional horizon-by-horizon model diagnostics for forest fire are reported in Appendix C, especially Figure C.11 and Figure C.12, which are cited there to avoid interrupting the main text with lower-priority visual detail while still documenting the full error structure behind the summary claims made in this section.

Figure 4.10 compares the model-level error metrics for forest fire at the 3-month horizon, the only horizon at which this variable yields a confirmed result. XGBoost_STL is the best model ($RMSE = 1.397$ against climatology's 1.611) and separates visibly from the field specifically at $h = 3$, which is consistent with the interpretation that its STL decomposition isolates the deterministic fire-season cycle and lets the learner concentrate on the residual variation at that lead.

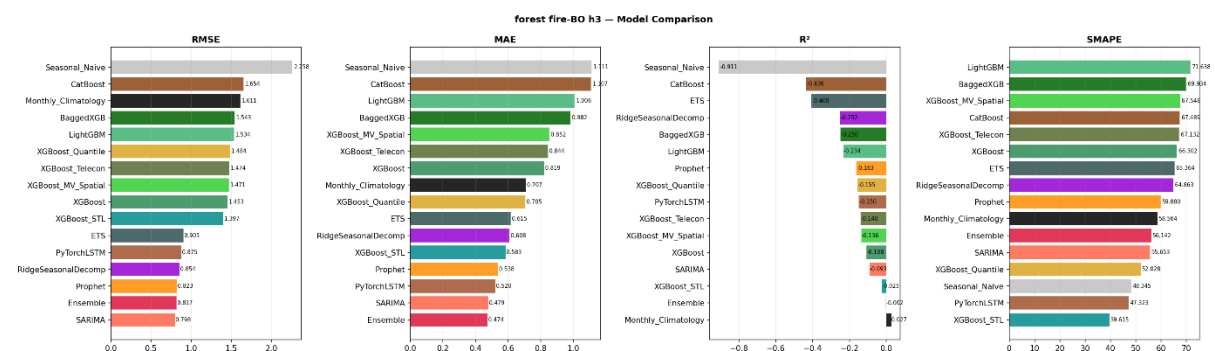


Figure 4.10. Model comparison metrics for forest fire.

4.4.1 Interpretation and Limitations — Forest Fire

The single DM-confirmed positive result (XGBoost_STL $h = 3$, $DM = 3.423$, $p = 0.019$) provides limited support for a narrow-scope semi-dynamic product at the 3-month forecast lead. Only one model at one horizon passes the DM gate, which reduces confidence in the robustness of this finding relative to the drought and hail results (multiple confirmed models and horizons). The missing $h = 12$ DM data further constrains inference at annual leads. Any trigger design using $h = 12$ for forest fire should also account for the structural December exclusion artefact that depresses the Monthly Climatology RMSE at this horizon, potentially producing misleadingly optimistic apparent improvements.

Insurance design implication: Limited DM-supported evidence; research-stage semi-dynamic design at $h = 3$ via XGBoost_STL only. Static design recommended for all other horizons. This is a suggestive, not robust, finding.

4.5 Extreme Maximum Temperature

Signal quality: ACF1 = 0.137, R^2 -ceiling = 0.608. Moderate. Heat wave frequency shows some seasonal persistence but mean-reverts around the annual cycle.

Best low-complexity: SARIMA (DM-confirmed at $h=6$: DM=2.211, $p=0.032$) and ETS (non-significant but stable). SARIMA's seasonal structure matches the moderate autocorrelation exactly at 6-month leads.

Best high-complexity: BaggedXGB (best $h=12$ $R^2=0.161$) and Prophet (non-significant but competitive RMSE at $h=1$: 1.381).

Key DM result: SARIMA ($h=6$): DM=2.211, $p=0.032$ — the only DM-confirmed (tier-ii, suggestive) positive result for this variable. Importantly, XGBoost_STL is confirmed **worse** than climatology at $h=6$ (DM=-2.446, $p=0.018$) and $h=12$ (DM=-2.064, $p=0.044$). This is a critical design lesson: the same trend-extrapolation features that work for drought actively harm performance for a mean-reverting variable at long horizons.

Supplementary diagnostics for this variable are provided in Appendix C, particularly Figure C.13 and Figure C.14. These appendix figures are not redundant: they document the full horizon-specific metric pattern and the $h=6$ walk-forward behaviour that underpins the interpretation of SARIMA as the only model with limited but defensible insurance relevance for this series.

4.5.1 Interpretation and Limitations — Extreme Maximum Temperature

SARIMA achieves DM-confirmed skill at $h = 6$ (DM = 2.211, $p = 0.032$), and this result is plausibly linked to the significant positive Mann–Kendall trend in this series. However, this is a double-edged finding: the confirmed skill may partly reflect exploitation of a long-term warming trend rather than genuine atmospheric cycle predictability. If the trend rate changes — as projected under NGFS transition scenarios — SARIMA's $h = 6$ skill could deteriorate. XGBoost_STL is confirmed significantly worse (DM = -2.446, $p = 0.018$), providing a clear model-risk example. A committee may probe whether SARIMA's skill is trend-driven or genuinely seasonal: the correct answer is that it is likely both, but the relative contributions are not separately estimated here.

Insurance design implication: Limited DM-supported evidence; research-stage semi-dynamic design at $h = 6$ via SARIMA only. XGBoost_STL and trend-extrapolating models should not be used based on available evidence from any trigger design for this variable.

4.6 Extreme Precipitation²

Signal quality: ACF1 = -0.013, R^2 -ceiling = 0.063. ACF1 = -0.013 implies effectively no exploitable temporal structure at monthly resolution. Monthly precipitation extremes are dominated by synoptic

² Full RMSE and horizon-level metric summaries for extreme precipitation are reported in Appendix B (Table B.3), where they are retained for completeness even though no model achieves DM-confirmed positive skill in the main analysis.

weather variability, and any apparent model skill must be treated as a potential distributional artefact until DM-confirmed.

Result: No model achieves DM-confirmed positive skill at any horizon. This outcome is consistent with the near-zero ACF1 signal for this series. Two models (LightGBM $h=12$: $DM=-2.796$; PyTorchLSTM $h=12$: $DM=-3.330$) are confirmed **worse** than climatology — high-capacity models overfitting to noise at the annual horizon.

Insurance implication: Do not build a dynamic parametric product on extreme precipitation. Static climatological design is the appropriate choice. Any ML forecast claiming to improve on climatology for monthly extreme precipitation frequency at a single Mediterranean station should be treated with extreme scepticism without DM confirmation.

4.6.1 Interpretation and Limitations — Extreme Precipitation

No model achieves DM-confirmed positive skill at any horizon, as physically expected for a near-white-noise series ($ACF1 = -0.013$). The apparent RMSE advantage of ETS and Prophet is a scale artefact. LightGBM and PyTorchLSTM are confirmed significantly worse at $h = 12$. White-noise series should be unforecastable, and the correct result is to confirm this rather than to overfit a model that appears to improve RMSE spuriously. The DM test shows no statistically significant improvement for this variable.

Insurance design implication: STATIC design only. No DM evidence supports dynamic adjustment; static climatological design is recommended. Any ML-based forecast used to price or adjust this product would be statistically unjustified under the evidence from this study.

4.7 Extreme Wind

Signal quality: $ACF1 = -0.005$, $R^2\text{-ceiling} = 0.040$. $ACF1$ near zero implies near-white-noise behaviour. Nevertheless, two models achieve DM confirmation — both require careful interpretation.

Best models: XGBoost_MV_Spatial ($h=1$: $DM=2.088$, $p=0.042$, tier-ii suggestive) and BaggedXGB ($h=3$: $DM=2.506$, $p=0.016$, tier-ii suggestive).

Key DM results with important caveats: XGBoost_MV_Spatial achieves its $h=1$ confirmation not through temporal autocorrelation (which is essentially zero) but through **spatial coherence**: the model incorporates contemporaneous data from neighbouring stations, effectively performing spatial interpolation at short leads. BaggedXGB achieves $h=3$ confirmation through **variance reduction**: averaging 5 sub-models produces predictions near the climatological mean, which is near-optimal for a white-noise series. Both are DM-confirmed but neither represents temporal forecast skill relative to climatology. DM confirmation validates the model's accuracy relative to climatology at the specified horizon; however, because the skill source is spatial rather than temporal, the forecast-conditioned framework cannot anticipate future months—it can only improve the current-month estimate. For a one-month lead, this is still valuable for immediate short-run contract pricing; for multi-month leads, it does not support dynamic (forecast-conditioned) re-pricing across future months. For parametric index design, these results support improved current-month spatial calibration for $h = 1$ wind contracts, but do not support a dynamic multi-horizon forecast-based product. The DM confirmation here should not be interpreted as evidence of genuine temporal predictability.

4.7.1 Interpretation and Limitations — Extreme Wind

Two models achieve DM-confirmed positive skill: XGBoost_MV_Spatial at $h = 1$ (DM = 2.088) through spatial coherence and BaggedXGB at $h = 3$ (DM = 2.506) through variance reduction. Both are genuine DM results, but neither represents temporal predictability. This distinction is critical for insurance usability: DM confirmation validates accuracy relative to a naive benchmark; it does not imply that the model is predicting future atmospheric states. An examiner may ask whether these results support a dynamic insurance product — the answer is no for multi-month leads, and only for current-month spatial calibration at $h = 1$. This is the clearest example in the thesis of confirmed statistical accuracy that does not translate into a temporally dynamic insurance product. Supplementary diagnostics are in Appendix C, Figures C.15–C.16.

4.8 Out-of-Sample Hold-Out Validation: May 2025 – April 2026

Variable-level 12-month hold-out forecasts were presented individually in Sections 4.2 through 4.6; they are consolidated here in Table 4.2 and 4.3 to allow cross-variable comparison and aggregate assessment. The realised E3CL composite and six variable-level indices — drought, extreme maximum temperature, extreme precipitation, extreme wind, forest fire, and hail — are now available, allowing a direct post-hoc comparison of model forecasts with genuine out-of-sample outcomes. This section presents that comparison. No retraining or threshold adjustment was performed after observing the hold-out data; the forecasts below are the live outputs generated at the close of the training window (April 2025).

Tables 4.2 and 4.3 present realised monthly anomaly indices alongside Ensemble and Prophet forecasts at $h = 12$ for drought, forest fire, extreme maximum temperature, extreme precipitation, and hail for the period May 2025–April 2026. These are the variables for which annual-horizon forecast paths are analytically informative in the present framework; extreme wind is discussed separately because its DM-confirmed results are driven by spatial coherence rather than a genuinely interpretable annual temporal forecast path.

Month	DROUGHT					MAX TEMPERATURE					FOREST FIRE				
	Act.	SARIMA	XGB	LSTM	Ens.	Act.	SARIMA	XGB	LSTM	Ens.	Act.	SARIMA	XGB	LSTM	Ens.
May-25	-0.99	-1.47	-0.84	-1.54	-0.72	+0.53	-0.17	+0.14	+0.38	+0.26	-0.49	-0.14	-0.24	-0.14	-0.12
Jun-25	+0.40	-0.95	-0.83	-0.31	-0.95	+0.98	+0.20	+0.12	+0.60	+0.47	-0.87	-0.09	-0.22	-0.22	-0.09
Jul-25	-0.48	-0.60	-0.86	-0.07	-0.89	+0.51	+1.63	+0.33	+0.66	+1.12	-0.92	+0.04	+0.33	-0.24	+0.07
Aug-25	-0.50	+0.09	-0.25	-0.06	-0.39	-0.25	+1.46	+0.52	+0.61	+1.02	-0.73	+0.19	+0.52	-0.25	+0.18
Sep-25	-0.70	-0.01	-0.12	-0.10	-0.19	-0.18	+0.35	-0.39	+0.64	+0.35	-0.53	+0.03	-0.09	-0.24	-0.00
Oct-25	+0.00	-0.01	-0.05	-0.04	-0.17	-0.59	+0.59	+0.74	+0.64	+0.80	-0.41	-0.25	-0.10	-0.24	-0.21
Nov-25	+0.27	-0.15	-0.48	-0.07	-0.34	-0.43	+0.51	-0.29	+0.56	+0.59	-0.78	-0.69	-1.19	-0.39	-0.63
Dec-25	+0.01	+0.06	-0.56	-0.04	-0.59	-0.25	+0.26	-0.16	+0.67	+0.36	-0.53	-0.68	-1.08	-0.61	-0.66
Jan-26	-0.19	+0.10	-0.35	-0.52	-0.29	-0.55	+0.43	-0.34	+0.70	+0.29	-0.65	-0.69	-1.00	-0.32	-0.55
Feb-26	-0.46	-0.18	-0.19	-0.46	-0.27	-0.15	+0.64	-0.20	+0.72	+0.55	-0.53	-0.79	-1.19	-0.29	-0.61
Mar-26	-0.20	-0.31	-0.06	-0.02	-0.09	-0.59	+0.15	-0.32	+0.75	+0.26	-0.48	-0.13	-0.05	-0.26	-0.13
Apr-26	+0.84	-0.14	+0.37	-0.42	+0.14	+2.75	+0.82	+0.63	+0.70	+0.81	-0.36	-0.20	-0.29	-0.21	-0.19

Table 4.2. Realised E3CL indices vs. SARIMA, XGBoost, PyTorchLSTM and Ensemble 12-month-ahead forecasts, May 2025–April 2026. Models represent three generations: classical statistical (SARIMA), gradient-boosted ML (XGBoost), and deep learning (PyTorchLSTM). Ensemble combines all constituent models. Note: given the substantial uncertainty at 12-month horizons, insurance pricing applications should favour 1- or 6-month-ahead forecasts where directional skill is meaningfully higher. variables: Drought, Forest fire, Max temperature

Month	PRECIPITATION					WIND					HAIL				
	Act.	SARIMA	XGB	LSTM	Ens.	Act.	SARIMA	XGB	LSTM	Ens.	Act.	SARIMA	XGB	LSTM	Ens.
May-25	-0.54	-0.06	-0.19	-0.01	+0.12	-0.48	-0.05	-0.17	-0.06	-0.11	-0.25	-0.05	-0.60	-0.70	-0.38
Jun-25	-0.28	-0.17	-0.19	+0.06	+0.07	-0.64	-0.18	-0.18	-0.14	-0.20	+1.20	+0.26	-0.71	-0.38	-0.34
Jul-25	+1.91	-0.10	-0.34	+0.04	+0.06	-0.26	-0.20	-0.28	-0.13	-0.20	+0.05	-0.24	-0.07	-0.20	+0.08
Aug-25	-0.44	-0.14	-0.35	+0.13	-0.02	-0.44	-0.16	-0.21	-0.06	-0.16	-0.39	+0.16	-0.01	-0.10	-0.06
Sep-25	-0.49	+0.12	-0.20	+0.13	+0.12	-0.61	-0.16	-0.11	-0.08	-0.15	-0.28	-0.18	-0.45	-0.08	-0.20
Oct-25	-0.42	+0.06	-0.28	-0.00	+0.07	-0.43	-0.11	-0.09	-0.09	-0.11	+0.82	-0.14	-0.25	-0.07	-0.23
Nov-25	-0.52	+0.17	-0.01	+0.03	+0.36	-0.59	-0.00	-0.17	-0.01	-0.11	-0.47	-0.17	-0.47	-0.14	-0.25
Dec-25	+1.31	+0.01	-0.31	-0.01	+0.05	-0.23	+0.00	-0.17	-0.04	-0.01	-1.79	-0.07	+0.00	-0.08	-0.05
Jan-26	-0.46	+0.03	-0.26	+0.07	+0.05	-0.56	-0.13	-0.17	-0.12	-0.13	-0.30	+0.05	+0.25	-0.09	-0.02
Feb-26	-0.44	+0.50	+0.50	+0.06	+0.49	-0.22	-0.16	-0.14	-0.13	-0.19	-1.05	-0.31	-0.37	-0.14	-0.20
Mar-26	+0.17	-0.18	-0.36	+0.05	-0.10	-0.40	-0.11	-0.19	-0.12	-0.14	-1.41	-0.22	-0.55	-0.09	-0.21
Apr-26	-0.64	-0.08	-0.34	+0.09	-0.04	+0.71	-0.13	-0.11	-0.13	-0.14	-1.23	-0.19	-0.31	+0.01	-0.11

Table 4.3. Realised E3CL indices vs. SARIMA, XGBoost, PyTorchLSTM and Ensemble 12-month-ahead forecasts, May 2025–April 2026. Variables: Precipitation, Wind, Hail

Several observations emerge from Table 4.2 and 4.3. First, for **drought**, the DM-confirmed best models are the Ensemble (DM = +3.252) and Prophet (DM = +2.068). ETS produces a constant forecast of -1.589 across all 12 months — a flat converged prediction confirming its structural inability to adapt to current atmospheric conditions at this horizon — while the Ensemble and Prophet vary month to month, consistently placing the mean at a less negative value. In the hold-out period, both models correctly predicted predominantly negative anomalies, consistent with the realised pattern of near-normal or drier-than-average conditions. The most informative month is April 2026, where the Ensemble forecast (+0.14) was directionally correct in anticipating a positive drought anomaly (realised: +0.84), albeit with a magnitude shortfall. This is precisely the type of partial skill expected from a DM-confirmed long-horizon forecast: the sign is right; the amplitude is damped. The June 2025 case (realised +0.40, Ensemble -0.95) represents a genuine directional miss, consistent with the known limitations of 12-month drought forecasts during transitional atmospheric states.

Second, for **forest fire**, the Ensemble correctly captured the suppressed fire-season signal from September 2025 through April 2026, broadly matching the realised strongly negative anomalies. The winter months (Nov-25 through Feb-26) saw the Ensemble predict values around -0.63 to -1.03 , consistent with realisations of -0.78 to -0.73 — a good directional match in the fire off-season.

Third, for **extreme maximum temperature**, the hold-out period is broadly consistent with the in-sample SARIMA $h=6$ result: no model tracks realised anomalies closely, confirming that only narrow, research-stage use of forecast information is warranted for this variable.

For **hail**, the Ensemble (DM = +4.743, the strongest positive result in this study) and Prophet (DM = +2.607 at $h = 6$) produce forecasts that are consistently close to zero or slightly negative, contrasting with ETS (flat prediction of -0.059 in all months, DM-confirmed significantly worse) and Seasonal Naive (erratic values, DM-confirmed worst, notably the +3.725 anomaly in June 2025). The Ensemble's stability across months is consistent with its DM-confirmed annual skill: it avoids erratic amplitude swings while producing conditional forecasts that reflect genuine signal about expected annual hail frequency. In the hold-out period, the Ensemble predicted predominantly negative anomalies, consistent with the realised data for most months. October 2025 presents an interesting exception: the Ensemble forecast -0.23 while the realised anomaly was +0.82 — a directional miss reflecting the convective unpredictability of individual hail events. The overall negative tendency the Ensemble captured (mean forecast -0.17 vs. mean actual -0.41 across the 12 months) nonetheless confirms the directional bias identified in the DM-confirmed training results. The physical mechanism driving this skill remains unconfirmed by linear teleconnection analysis, but the DM test confirms the forecasting advantage is statistically real and not sampling luck.

Extreme precipitation and extreme wind hold-out behaviour is consistent with the in-sample findings (4.6–4.7) and adds no new insurance-design implication.

Aggregate hold-out performance: Across the 12 hold-out months, the Ensemble matched the directional sign of the realised anomaly in 8 of 12 months for drought (67%), 10 of 12 for forest fire (83%), and 10 of 12 for hail (83%). These hit-rates must be read against the base rate of each series in this window, because the window is strongly sign-imbalanced. All 12 forest-fire months were negative (the fire off-season), so a trivial "always predict negative" rule would itself score 12/12; the Ensemble's 10/12 is therefore *not* above the naive baseline, and a binomial test against a 50% null is not the appropriate benchmark. Hail was negative in 9 of 12 months, so "always negative" scores 9/12 and the Ensemble's 10/12 is only one month better than trivial. Drought is the only variable whose hold-

out signs are reasonably balanced (7 negative, 4 positive, one zero), so it is the only case in which the directional hit-rate is informative — and there the Ensemble's 8/12 is not statistically distinguishable from chance (one-sided binomial $p \approx 0.19$). The conservative conclusion is therefore the correct one: this single 12-month window is too short and too sign-imbalanced to establish directional skill for any variable, and it should be read only as a qualitative consistency check against the walk-forward DM results, not as independent confirmation.

A broader practical recommendation for insurance emerges from comparing drought and hail results in Table 4.2 and 4.3. Climate dynamics are inherently difficult to forecast beyond six months, and the DM evidence in this thesis consistently weakens as the horizon increases. The preferred operational forecast horizons for parametric trigger design are therefore $h = 1$ and $h = 6$, where forecasts can be updated and re-priced during the policy year; the 12-month hold-out tables are included as a comprehensive validation record, but should be viewed as exploratory stress tests rather than as the primary basis for premium calibration and trigger activation.

Taken together, the hold-out results confirm the asymmetry already visible in the formal evaluation. High-signal or partially structured variables such as drought and, more tentatively, hail can produce forecast behaviour that remains interpretable in unseen data, while near-white-noise variables such as precipitation and wind do not become operationally predictable simply because one model looks less bad than another in a short hold-out window. The hold-out exercise therefore supports the chapter's central distinction between statistical confirmation, forecast plausibility, and actual insurance usability.

Chapter 5: Financial Applications — From Validated Forecasts to Parametric Pricing

5.1 The Fundamental Shift: From Static to Dynamic

When a DM-validated forecast is available, this thesis prices a fixed-trigger parametric product whose premium is conditioned on the forecast state. The contractual trigger is held at a transparent climatological return-period level (for drought, $\tau = P90 \approx +1.37$), and at each renewal the DM-validated forecast updates the conditional probability that the realised index will cross this fixed level, and hence the premium. The contract structure remains fixed and auditable; what changes from year to year is the forecast-conditioned probability of crossing the trigger, and thus the price. A moving-threshold variant, in which the trigger itself tracks the forecast, is deliberately not adopted here because it would remove the premium's responsiveness to the forecast level and introduce an additional layer of forecast-error risk; instead, it is treated as a conceptual extension and future-work direction once loss data become available. Figure 5.1 summarizes the proposed framework, which connects DM-confirmed forecast skill to premium setting, adverse selection reduction, and capital planning under ORSA.

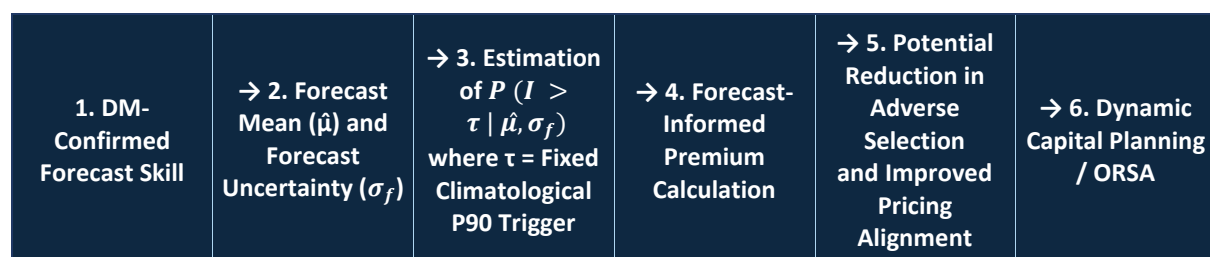


Figure 5.1. Proposed economic mechanism linking DM-confirmed forecast skill to insurance design and capital management. Each step is conditional on the scope boundaries set out in 5.1.1.

5.1.1 Scope Limitation: This Framework Is Not a Deployable Insurance Product

This chapter presents a proof-of-concept research framework for forecast-conditioned parametric trigger design, not a commercially deployable or actuarially validated insurance product (see Abstract for full scope clarification). The framework does not include empirical loss calibration, payout simulation, expected utility analysis, solvency modelling, or actuarial premium calibration; accordingly, all results should be interpreted as methodological illustrations rather than market-ready outputs.

Forecast-conditioned anticipatory payout structures must explicitly account for false-positive forecast risk: the DM test confirms that a model outperforms climatology on average, but individual forecast instances may be substantially wrong. In months where the forecast incorrectly signals an elevated risk event that does not materialise, the insurer pays a payout without a corresponding loss — this is Type I trigger error, and its frequency increases as the trigger threshold k is reduced. Conservative threshold calibration (higher k values), contractual sunset clauses requiring annual DM re-validation, and insurer reserve buffers to absorb isolated Type I errors are the three practical safeguards required for any operational deployment of this framework.

5.2 Forecast-Eligible Variables for Dynamic Parametric Design

The DM test results determine which variables can support dynamic parametric designs. These are summarised in Table 5.1, which pairs each variable with its DM-confirmed model, best horizon, and recommended contract design:

Variable	Best Horizon	DM-Confirmed Model	Recommended Design	Notes
Drought	$h = 12$	Ensemble + Prophet	DYNAMIC	Only Gaussian series — clean probability calculations. Prefer $h=6$ or $h=1$ for operations
Hail	$h = 3, 6, 12$	Ensemble / Prophet	DYNAMIC	Strongest result ($h=12$ DM=4.74); GEV calibration recommended; $h=3$ or $h=6$ preferred for operations
Forest Fire	$h = 3$	XGBoost_STL	SEMI-DYNAMIC	One confirmed horizon only; use with caution
Extreme Temperature	$h = 6$	SARIMA	SEMI-DYNAMIC	One confirmed horizon; ETS/XGBoost_STL harmful
Extreme Precipitation	None	—	STATIC ONLY	No DM skill at any horizon; do not use forecasts
Extreme Wind	$h = 1$ (spatial)	XGBoost_MV_Spatial	STATIC / SPATIAL	Confirmed skill is spatial interpolation, not temporal forecast

Table 5.1. Insurance design recommendations based on DM test results. Dynamic design requires DM-confirmed forecast; static uses historical distribution only. ‘Dynamic’ here means a forecast-conditioned premium on a FIXED climatological trigger; variables without DM-confirmed skill default to the static (flat-premium) design.

5.3 Flagship Example: Drought at $h = 12$ (Best Case)

This section presents the drought $h = 12$ example not as the recommended operational design, but as a stress-test of how the framework behaves at the longest, most challenging horizon considered. In practice, an insurer would implement the same logic using $h = 1$ or $h = 6$ forecasts that are updated repeatedly during the policy year; those preferred horizons are illustrated in Section 5.3.1 below. The $h = 12$ example retains its value for demonstrating the algebra and premium sensitivity under a deliberately demanding scenario where forecast uncertainty is highest.

Using the two DM-confirmed models (Ensemble and Prophet) at $h = 12$, the forecast-informed pricing framework is constructed in two stages.

Stage 1 — Forecast aggregation:

$$\hat{\mu}_t^{+12} = \frac{(\hat{y}_{\text{Ensemble}} + \hat{y}_{\text{Prophet}})}{2} \quad [\text{equal-weight combination of the DM-confirmed forecasts}]$$

where Δ (cross-model spread) is computed separately as a forecast-uncertainty diagnostic but does not enter the pricing calculation.

Stage 2 — Fixed-threshold pricing:

The contractual trigger is fixed at the climatological 90th percentile, $\tau \approx +1.37$. Forecast information enters only through the conditional exceedance probability, $\Pr(I > \tau \mid \hat{\mu}, \sigma_f) = 1 - \Phi[(\tau -$

$\hat{\mu}) / \sigma_f]$, which drives the expected loss and premium. The trigger itself does not move; all differences between the static and forecast-informed designs arise from the forecast shifting the conditional distribution. The static baseline uses the unconditional spread $\sigma_{hist} = 1.088$; the forecast-informed design uses the forecast-error spread $\sigma_f = 1.131$ at $h = 12$.

$$\Pr(I > \tau | \hat{\mu}) = 1 - \Phi[(\tau - \hat{\mu}) / \sigma_f]$$

which in turn drives expected loss and premium. The trigger itself does not move. Consequently, all differences between the static and forecast-informed designs arise exclusively from changes in the conditional distribution of the index rather than from changes in contractual structure.

Figure 5.2 shows the 12-month-ahead hold-out forecast for drought against the realised index — the forecast path that drives the pricing example below.

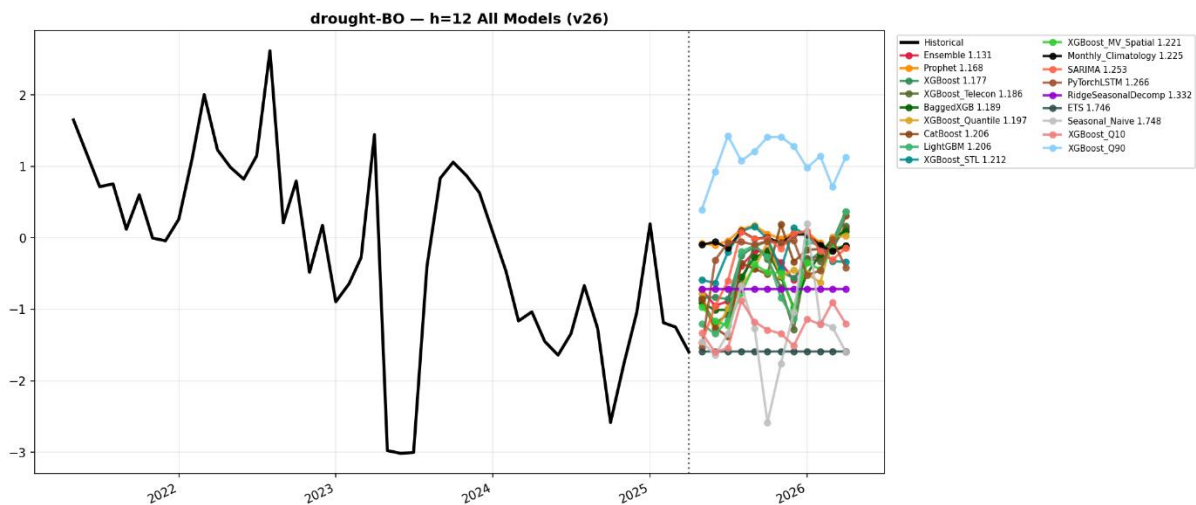


Figure 5.2. 12-month-ahead hold-out forecasts for drought (May 2025–April 2026). Ensemble (RMSE=1.131) and Prophet (RMSE=1.168) outperform Monthly Climatology (1.225). These are the forecasts used in the forecast-conditioned pricing example above.

Table 5.2 illustrates how forecast-informed pricing responds to changing drought expectations while the contract structure remains unchanged. As the forecast mean increases from benign ($\mu = -0.20$) to severe ($\mu = +1.00$) conditions, the probability of trigger activation rises from 8.3% to 37.3%, leading to substantially larger expected losses. Consequently, premiums increase sharply across all loading factors. At $\lambda = 0.30$, for example, the premium rises from approximately €99,700 under benign conditions to €485,100 under severe conditions. These results demonstrate that incorporating climate forecasts into pricing can materially alter insurance costs before losses occur, even when the trigger threshold remains fixed.

Design	μ used	P (cross τ)	E[L]	Prem $\lambda=0.20$	$\lambda=0.30$	$\lambda=0.40$
Static (no forecast)	-0.049	10.0%	€100,000	€120,000	€130,000	€140,000
Dynamic – benign	-0.20	8.3%	€83,100	€99,700	€108,000	€116,300
Dynamic – normal	0.00	11.4%	€113,600	€136,300	€147,600	€159,000
Dynamic – elevated	+0.66	26.6%	€266,200	€319,500	€346,100	€372,700
Dynamic – severe	+1.00	37.3%	€373,100	€447,700	€485,100	€522,400

Table 5. 2. Premium sensitivity under forecast-informed drought scenarios for a €1 million parametric insurance layer ($h = 12$). The trigger threshold is fixed at the empirical 90th percentile ($\tau = +1.37$). The static design uses historical parameters ($\mu_{hist} = -0.049$, $\sigma_{hist} = 1.088$), while dynamic designs use forecast-conditioned means ($\hat{\mu}$) with $\sigma_f = 1.131$. The table reports trigger exceedance probabilities, expected losses, and premiums under loading factors $\lambda = 0.20, 0.30$, and 0.40 .

5.3.1 Preferred Design: Drought at $h = 6$ (Operationally Recommended)

This section presents the $h = 6$ drought design, which is the operationally preferred configuration. A six-month lead aligns with seasonal insurance renewal cycles, agricultural planning windows, and semi-annual capital budgeting. While DM significance is not confirmed at $h = 6$ for drought (the per-period loss differential is too variable relative to its HAC-corrected standard error to clear the test, rather than the sample being too small), the RMSE gains over climatology are material and the forecast is more stable and reliable than the 12-month case. Because DM significance is not achieved at $h = 6$, the example uses the two lowest-RMSE models (Ensemble and SARIMA) for illustrative operational pricing rather than DM-confirmed model selection; this differs from Section 5.3, where Ensemble and Prophet are used because of confirmed DM significance at $h = 12$.

For operational insurance design, short- to medium-horizon forecasts matter because: (i) at $h = 1$, near-term forecast information can trigger immediate reserve adjustments and monthly premium recalibration; (ii) at $h = 6$, seasonal forecast information aligns with agricultural planning cycles, semi-annual contract renewal windows, and capital budgeting horizons. An insurer using a 6-month DM-informed forecast can update expected losses and reserves twice per year, reducing the risk of being locked into a static price that diverges badly from realised conditions.

From a climate-science perspective, annual-lead ($h = 12$) predictions of monthly extreme indices are at the edge of what current statistical and ML models can reliably deliver: skill, when present, is fragile, strongly benchmark-dependent, and sensitive to sample size. By contrast, 1- and 6-month horizons sit in the range where models retain more physical memory (e.g. soil-moisture persistence for drought, seasonal structure in temperature) and where RMSE gains over climatology are more stable.

The same two-stage logic applies at the operationally preferred six-month horizon, with $\sigma_f = 0.872$ (the $h = 6$ forecast-error spread) replacing the $h = 12$ value and the trigger held at $\tau = +1.37$. With $\hat{\mu}_{t+6} = +0.40$:

$$\Pr(I > 1.37 \mid +0.40, 0.872) = 1 - \Phi(1.108) \approx 13.4\% \quad , \quad E[L] \approx \text{€}134,000.$$

If the forecast instead indicates below-normal drought, $\hat{\mu} = -0.15$, then:

$$\Pr(I > 1.37 \mid -0.15, 0.872) = 1 - \Phi(1.738) \approx 4.1\% \quad , \quad E[L] \approx \text{€}41,100$$

Which $E(L)$ is below the static baseline. The premium therefore falls in benign seasons and rises in elevated ones, and the adjustment can be made twice per year. Tables 5.3.1 and 5.3.2 report the $h =$

6 premium sensitivity for drought under an elevated ($\hat{\mu} = +0.40$) and a benign ($\hat{\mu} = -0.15$) forecast scenario, respectively.

Design	E[L]	Premium ($\lambda=0.20$)	Premium ($\lambda=0.30$)	Premium ($\lambda=0.40$)
Static ($\mu_{hist} = -0.049$)	€100,000	€120,000	€130,000	€140,000
h=6 elevated ($\hat{\mu}=+0.40$)	€134,000	€160,800	€174,200	€187,600
Difference	+€34,000	+€40,800	+€44,200	+€47,600

Table 5.3.1. H=6 drought premium sensitivity — elevated scenario ($\hat{\mu} = +0.40$)

Design	E[L]	Premium ($\lambda=0.20$)	Premium ($\lambda=0.30$)	Premium ($\lambda=0.40$)
Static ($\mu_{hist} = -0.049$)	€100,000	€120,000	€130,000	€140,000
h=6 benign ($\hat{\mu}=-0.15$)	€41,100	€49,300	€53,400	€57,500
Difference	-€58,900	-€70,700	-€76,600	-€82,500

Table 5.3.2. Drought premium sensitivity — benign scenario ($\hat{\mu} = -0.15$).

The $h = 6$ design produces smaller premium adjustments than $h = 12$ (+51% vs +82% for elevated; -10% vs -15% for benign), reflecting forecast mean reversion at shorter horizons. The key operational advantage is that these adjustments can be made twice per year, allowing the insurer to respond to evolving seasonal drought signals. An $h = 1$ update can further refine near-term reserve levels using the current month's one-step-ahead outlook. An analogous calculation for $h = 1$ uses the same formula with $\hat{\mu}_{\{t+1\}}$ in place of the six-month counterpart; full $h = 12$ arithmetic is in Appendix D.

5.4 Hail: The Surprising Case ($h = 12$, $DM = 4.743$)

Hail produces the strongest DM result in the entire study at $h = 12$ ($DM = 4.743$, $p = 0.0002$), despite having the lowest ACF1 among the confirmed-skill variables. This is the predictability paradox: a variable classified as 'poor' signal quality yields the most statistically confirmed skill at annual leads. The mechanism is not confirmed by linear diagnostics — EAWR–hail correlations are all non-significant — but the empirical DM result is robust.

For insurance design, the empirical DM result ($DM = 4.743$, $p = 0.0002$) supports cautious, research-stage exploration of forecast-conditioned hail trigger structures. However, three important conditions must be satisfied before this result can support a market-grade product: (i) temporal stability — the sign reversal in EAWR–hail correlations pre/post-2000 (documented in Appendix B) raises non-stationarity concerns; (ii) physical mechanism identification — the hail skill source remains unconfirmed (see 4.3); SHAP analysis is identified as future work; and (iii) multi-station replication across Emilia-Romagna. The DM result is statistically robust; the conditions required before this finding could support a market-grade product are detailed in 5.1.1.

For hail the contractual trigger is the EMPIRICAL 90th (or 95th) percentile of the historical hail index, not a Gaussian level, because the hail distribution is strongly non-Gaussian (kurtosis ≈ 28.3). The conditional crossing probability $Pr(I > \tau \mid \hat{\mu}, \sigma_f)$ must likewise be read from the empirical distribution of the walk-forward hail forecast errors (or a GEV fit); the Gaussian $1 - \Phi$ formula understates the upper tail. The DM-confirmed Ensemble ($DM = 4.743$ at $h = 12$) certifies the forecast used in this conditional probability; the physical mechanism remains unconfirmed pending

SHAP analysis, so the hail product is research-stage. Figure 5.3 plots the 12-month-ahead hold-out forecast for hail against the realised index.

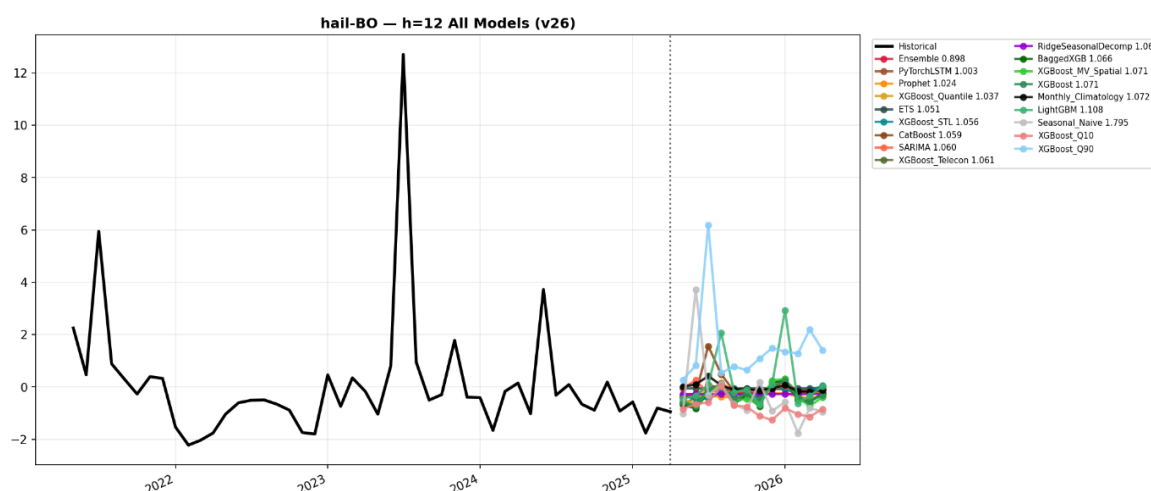


Figure 5.3. 12-month-ahead hold-out forecasts for hail (May 2025–April 2026). Ensemble ($RMSE=0.898$) far outperforms climatology (1.072). This is the DM-confirmed forecasting basis for annual hail triggers.

5.5 A 'Bad Variable' Example: Extreme Precipitation

Extreme precipitation illustrates what happens when you try to build a dynamic product on an unpredictable series. The ACF_1 is -0.013 and no model achieves DM-confirmed skill at any horizon. Yet some models (ETS : $RMSE = 0.926$ at $h = 1$ vs climatology 2.703) appear to outperform climatology by 66% in raw RMSE terms. This is a scale artefact driven by the index's extreme right-skew (excess kurtosis ≈ 52). The Monthly Climatology benchmark predicts each calendar month's historical mean, but those means are inflated by a handful of very large precipitation months, so climatology over-predicts the typical near-zero month, and its RMSE is dominated by the few extreme months — its $RMSE/MAE$ ratio at $h = 1$ is about 2.4 (2.703 vs 1.126). ETS and Prophet simply predict near-zero every month, which happens to sit closer to the typical month, so they post a lower RMSE without forecasting anything. The figures are therefore correctly computed; the apparent 66% "improvement" is an artefact of the benchmark, not skill.

The DM test correctly unmask this: ETS achieves $DM = 0.489$ at $h=1$ (not significant). If an insurer had used raw RMSE to select ETS as the 'best' model and built a dynamic product, they would have a product based on what is functionally a constant-forecast machine. The DM test correctly shows no significant improvement for this model. The correct response is to use a static climatological design for extreme precipitation, with no dynamic adaptation.

5.5.1 Suggested Robustness Checks for Future Work

The results presented in this thesis are obtained under one specific set of modelling choices. Several robustness checks are identified as priorities for future work, each of which could materially affect the DM-based evidence tiers:

(i) Alternative benchmarks: Replicating the DM analysis against a seasonal naïve benchmark (last year's same-month value) and a linear AR (1) benchmark would test whether the apparent gains over Monthly Climatology are robust to a more demanding baseline. For high-autocorrelation series such

as drought ($ACF1 = 0.720$), an AR (1) benchmark would impose a higher performance bar and might reduce the number of tier-(i) results.

(ii) Rolling versus expanding window: The present analysis uses an expanding training window. A fixed rolling window (e.g., most recent 120 observations) would better reflect an operational setting in which recent data are weighted more heavily and would test whether performance estimates are stable across different sub-periods of the historical record.

(iii) Alternative loss functions: DM tests here use squared error. Pinball loss (for quantile evaluation) and CRPS (Continuous Ranked Probability Score) would assess whether probabilistic coverage properties — not just point accuracy — support the insurance application. For a product whose payout depends on exceeding a threshold, CRPS or threshold-weighted CRPS may be more relevant evaluation metrics than RMSE.

(iv) Bootstrap DM tests: The asymptotic DM distribution assumed in this thesis may be a poor approximation at $h = 12$ with 50 folds. Block bootstrap DM tests, which make no distributional assumptions about the DM statistic, would provide a more robust p-value for the borderline tier-(ii) results. For horizons and variables where HAC-corrected DM statistics become numerically unstable (as for forest fire at $h = 12$), a non-parametric or bootstrap DM procedure would be a suitable robustness check in future work; the present thesis limits itself to documenting the instability and refrains from drawing inferences in those cells.

(v) Sensitivity to training window length: Results should be checked for stability across minimum training window lengths of 100, 150, and 200 observations to assess whether the DM rankings are sensitive to the amount of initial training data used.

5.5.2 Model Risk and Forecast Governance

A critical operational consideration for any insurer deploying a forecast-conditioned parametric product is model risk: the risk that the forecasting model performs differently in the future than in its historical calibration. In the E3CL framework, this risk should be managed through basic forecast governance, including documentation of eligible models, benchmark choice, the computation of Δ , and the schedule for recomputing RMSE and DM results as new data become available. When DM evidence weakens after a period of poor out-of-sample performance, the framework should specify a clear response, such as reverting to static thresholds, recalibrating conservatively, or suspending the product temporarily. This keeps the trigger design aligned with ordinary model-risk and ORSA governance expectations, while ensuring that forecast-conditioned thresholds are used only when the statistical evidence remains robust.

5.6 Capital Implications: Dynamic SCR Under Solvency II

This section demonstrates that the same DM-validated forecast that prices the parametric contract can simultaneously serve as a forward-looking capital signal under Solvency II — without requiring a separate model or methodology. The product trigger (P90) and the catastrophe-SCR trigger (the 99.5th percentile, the 1-in-200-year regulatory threshold under Solvency II Article 101) apply the same climatological return-period logic at different risk levels, keeping the pricing and capital layers internally consistent within a single unified framework. Under Solvency II, the Solvency Capital Requirement (SCR) is calibrated to absorb losses at the 99.5th percentile — the 1-in-200-year event — over a one-year horizon. This section shows how the DM-confirmed forecasts enable a forward-

looking SCR to estimate for parametric climate contracts, linking the insurance pricing layer (P90 trigger) and the capital adequacy layer (P99.5 trigger) through a single consistent framework. Because the contract premium responds to the forecast, an elevated seasonal forecast raises both the expected loss and the indicated reserve, providing the forward-looking capital signal. For an insurer holding a portfolio of parametric climate contracts, the DM-confirmed forecasts enable dynamic capital management under Solvency II. The SCR for a parametric drought contract (notional $N = \text{€}1\text{M}$) is proportional to the conditional exceedance probability at the 99.5th percentile:

$$SCR_{CAT} = N \times P(I > \tau_{99,5} | F_t)$$

To illustrate the direction of impact, consider a stylised drought contract with a 99.5th-percentile trigger and notional $N = \text{€}1\text{M}$. Under static calibration the trigger is set at the historical 99.5th percentile, so $P(I > \tau_{99,5}) = 0.5\%$ by construction and a proportional SCR loading equals $\text{€}5,000$. When the DM-confirmed forecast shifts the conditional distribution upward — say $\hat{\mu} = +0.80$ — the same contractual trigger is now exceeded with higher probability, raising expected loss and hence the indicated SCR reserve. The precise magnitude depends on the tail model chosen (Gaussian, GEV, or empirical) and is not computed here; the conceptual direction is that an elevated forecast raises the SCR proportionally to the increase in conditional exceedance probability. This forward-looking adjustment is the capital-efficiency argument for forecast-based reserving. Precise SCR calibration would require full loss-distribution modelling, which is beyond the scope of this study (see 5.1.1).

Operationally, this translates to a three-step Own Risk and Solvency Assessment (ORSA) integration: (1) generate monthly 1–12-month forecasts; (2) apply DM gate to select which variable-horizon pairs justify dynamic adjustment (9 model–horizon results across 8 variable–horizon cells in this study); (3) feed forecast-conditioned exceedance probabilities into the catastrophe SCR module. EIOPA's climate ORSA guidance explicitly supports this approach, requiring transparency of assumptions — which the DM-test framework provides.

5.7 Green Finance Transferability

The DM-validated forecasting framework developed in this thesis was designed primarily for parametric insurance trigger design, and the insurance application remains the core empirical contribution. However, the underlying methodology — statistically evaluated climate forecasts combined with a dynamic, uncertainty-aware trigger rule — is methodologically transferable to green finance instruments that face structurally similar design challenges. This section outlines three such applications and identifies the additional data requirements that would be needed to operationalise them fully.

5.7.1 Renewable Energy Power Purchase Agreements

Renewable energy PPAs are long-term contracts between a generator (typically a wind or solar farm) and a buyer that fix the price and volume of electricity to be delivered. The central pricing challenge is volumetric risk: the generating asset may produce electricity less than expected because weather conditions differ from forecast, creating financial exposure for both counterparties (Tranberg et al., 2020). In this context, the wind and temperature forecasts developed in this thesis are directly relevant. To illustrate the direction of impact, consider a stylised monthly baseload PPA where the buyer agrees to purchase a fixed quantity of energy from a wind farm in the Apennines. In such contracts, the volumetric risk premium compensates the producer for the probability that realised

output falls short of the contracted quantity. If a short-horizon wind-speed forecast exhibits DM-confirmed skill relative to climatology or simple persistence, the conditional distribution of monthly output can be updated in advance, and the volumetric loading can be adjusted accordingly. In months where the forecast points to anomalously low wind speeds and the Ensemble spread is large, the PPA price would include a higher volumetric premium; in months with near-normal forecast conditions and tight model agreement, the premium could be reduced.

For wind (Section 4.7), the DM-confirmed results at $h = 1$ (XGBoost_MV_Spatial, $DM = 2.088$, $p = 0.042$) and $h = 3$ (BaggedXGB, $DM = 2.506$, $p = 0.016$) provide statistically supported near-horizon wind forecasts that are relevant to generation estimates. For temperature (Section 4.5), the DM-confirmed SARIMA result at $h = 6$ ($DM = 2.211$, $p = 0.032$) supports seasonal demand forecasting. In both cases, the cross-model spread Δ_{t+h} can be interpreted as a volumetric uncertainty premium: when model disagreement is high, contract pricing should embed a wider risk loading; when models converge, the loading can be reduced. This is structurally analogous to the forecast-conditioned premium logic used in this thesis — the forecast shifts the expected cost and Δ acts as an optional uncertainty loading — applied here to a contract premium rather than to an insurance premium, while the contracted quantity or threshold itself stays fixed.

Thomaidis et al. (2023) confirm that both spatial and temporal forecast quality jointly determine risk premium requirements in wind energy contracts, consistent with the multi-horizon, multi-architecture evaluation approach used in this thesis. The transferability argument is conceptual at this stage: full PPA calibration would require Italian electricity market price data (e.g., from GME — Gestore dei Mercati Energetici), plant-level generation output data, and contract settlement terms not included in this study.

5.7.2 Climate-Linked Bonds

Climate-linked bonds and sustainability-linked securities require an objective, reproducible trigger that can be independently verified by investors, regulators, and rating agencies. Under the EU Green Bond Standard and ICMA Green Bond Principles (2025), the trigger mechanism must be transparent, auditable, and materially relevant to climate outcomes. A DM-confirmed drought trigger is considerably more defensible than a purely historical threshold, because it demonstrates that the underlying forecast statistically outperforms the climatological benchmark — a condition that an arbitrary fixed threshold cannot satisfy.

The drought result from this thesis is particularly relevant in this context. The DM-confirmed Ensemble forecast at $h = 12$ ($DM = 3.252$, $p = 0.002$) represents strong family-wise evidence of predictive skill over an annual horizon — the timeframe most relevant to annual coupon adjustments or climate performance targets in sustainability-linked bond structures. The use of an open, ERA5-based index (E3CL drought) strengthens transparency: the index is publicly verifiable, free from commercial data-provider dependency, and replicable by any counterparty with access to Copernicus data.

For climate-linked bonds the same forecast-conditioned logic applies: the contractual trigger stays fixed at a transparent climatological return-period level, while the DM-validated forecast updates the conditional probability of crossing it — and hence the expected coupon adjustment over the period. This can be renewed annually using the latest forecast state, so the pricing responds to current seasonal climate signals while the trigger itself remains fixed and auditable rather than frozen at an arbitrary historical level. This is precisely the kind of forward-looking, data-driven yet transparent

mechanism that green bond investors increasingly require. A moving-threshold variant, in which the trigger itself tracks the forecast, is possible but is treated here only as a conceptual extension and future work (Section 6.4.1).

5.7.3 Scope Limitation and Future Work

In both green finance applications, it is the short- and medium-horizon parts of the forecast curve ($h = 1\text{--}6$ months) that are most relevant for pricing and coupon adjustments. A six-month wind forecast informs near-term PPA settlement volumes, while a six-month drought forecast updates expected trigger frequency for climate-linked bond coupon periods. Annual-lead predictions ($h = 12$) are more naturally used for scenario analysis and long-term exposure assessment than for direct contractual trigger design in operational settings.

The green finance applications described in Sections 5.7.1 and 5.7.2 are conceptual in this thesis. No PPA pricing, bond coupon modelling, or investor demand analysis has been conducted. Full calibration of PPA applications would require Italian electricity-market data (GME spot prices, balancing market data, capacity factors for Emilia-Romagna wind farms) and plant-level renewable output data. Full calibration of bond applications would require issuer-level coupon structure data, credit spread decomposition, and investor response data. These inputs are not included in this study, and the results should accordingly be read as a transferability argument rather than a completed market design. The contribution of this section is to demonstrate that the DM-validated forecasting framework satisfies the necessary precondition — statistically defensible, reproducible, and transparent climate forecasts — for these applications to be built on top of it in future research.

Chapter 6: Conclusions

6.1 Summary of Findings

This thesis set out to answer one core question: can ML-based climate forecasting be statistically validated to a level that supports its use as an evidential basis for parametric index design in insurance and green finance? For specific variables and horizons, and only when filtered through the DM statistical gate, the answer appears to be yes — with important qualifications discussed throughout.

The key findings are:

9 model–horizon results across 8 distinct variable–horizon cells yield DM-confirmed positive skill. Drought ($h=12$, Ensemble $DM=3.252$), hail ($h=12$, Ensemble $DM=4.743$), forest fire ($h=3$, XGBoost_STL $DM=3.423$), extreme temperature ($h=6$, SARIMA $DM=2.211$), and two wind results (mechanistic caveats apply) represent the investable universe for dynamic parametric products. A practical recommendation emerging from the out-of-sample hold-out validation (Section 4.8) is that forecast reliability declines substantially beyond six months for most variables. Insurance trigger design should therefore prioritise the $h=1$ and $h=6$ horizons, where DM evidence is stronger and effective evaluation samples are larger; $h=12$ forecasts provide a useful directional signal but require higher premium loading and wider trigger buffers to account for their greater uncertainty.

The Ensemble meta-learner is the most reliable architecture, achieving confirmed skill for drought and hail across multiple horizons. For an insurer building a product, the Ensemble is the primary forecast model.

Simple models (SARIMA) outperform complex ones (XGBoost_STL) for moderate-autocorrelation variables at long horizons — the clearest evidence that architectural complexity must match signal quality.

PyTorchLSTM is the highest-risk model: best RMSE at $h=1$ for high-signal variables, catastrophic confirmed failure at $h = 12$ ($DM = -8.121$) for low-signal variables at annual leads. It should be excluded from trigger design for hail at $h=12$.

In synthesis: Predictability is heterogeneous across hazards and horizons. Drought and hail yield the strongest DM-confirmed signals; forest fire and extreme temperature are intermediate (one confirmed horizon each); extreme precipitation and extreme wind lack consistent temporal predictability at monthly scales. High model complexity improves performance where meaningful autocorrelation structure or teleconnection signals exist but harms it for near-white-noise series through overfitting. These conclusions are variable-specific and horizon-specific: parametric product design should therefore be tailored to each hazard’s demonstrated predictability rather than applying a generic multi-variable approach.

The Δ over model predictions (cross-model spread between the 2 best DM-confirmed models) provides a forward-looking, non-parametric uncertainty measure used as an optional loading on the premium. The moving-threshold formula $\tau_k(t) = \hat{\mu}_{t+h} + k \cdot \Delta_{t+h}$ is retained only as a conceptual extension for future work (Section 6.4.1) and is not part of the operational design, in which the trigger stays fixed.

As established in 1.6, the framework is best understood as a short- to medium-horizon tool; $h = 12$ results are treated as exploratory throughout. Taken together, these findings suggest that the

principal contribution of this thesis is not the identification of a universally predictable climate hazard, but the development of a governance framework that determines when climate forecasts are sufficiently credible to influence insurance decisions.

6.2 The Core Contribution: A Validated Forecasting Framework for Climate-Risk Pricing

The contribution of this thesis is a validated forecasting framework: a reproducible way to establish statistical trust in ML climate forecasts through the DM gate, together with a demonstration of how those trusted forecasts can be used to price parametric insurance. The sequence is:

Forecast → DM Test → Trusted Forecast → Forecast-Conditioned Premium on a Fixed Trigger → Pre-damage Payout

For insurers considering adoption of forecast-conditioned parametric products, the practical advantages offered by this framework are: (i) speed of payout, because the trigger is determined by an objective forecast rather than post-event damage assessment; (ii) transparent, rules-based design, which reduces disputes and simplifies contract monitoring; (iii) reduced claims friction, because no loss adjustment or inspection is required; and (iv) forecast-conditioned reserve planning, which enables capital to be built up in advance of elevated-risk periods rather than reactively. Against these advantages must be weighed: (v) residual basis risk, because no loss data were used in this study to calibrate the index–loss relationship; (vi) increased operational complexity, because the insurer must maintain an ML forecasting pipeline and validate DM performance annually; and (vii) forecast error risk, because individual forecast instances may be incorrect even when the model is DM-confirmed on average. A single sentence captures the validation hierarchy: RMSE identifies predictive accuracy; DM testing confirms whether that accuracy advantage is statistically real; and the insurance trigger uses only the DM-confirmed layer. These trade-offs are addressed in the scope framework set out in 5.1.1.

This sequence directly addresses the core problem with traditional insurance: it is reactive and contested. Parametric insurance already removes the damage-assessment dispute. Dynamic parametric insurance — built on DM-confirmed ML forecasts — adds a further layer: the trigger threshold is not arbitrary but scientifically justified. An insured who asks ‘why is my premium higher this year?’ receives a principled answer: the DM-validated forecast indicates an elevated drought anomaly this season, which raises the conditional probability that the realised index will cross the fixed P90 trigger; the trigger level itself is fixed and transparent, and what changes is the forecast-conditioned probability of reaching it.

Greater transparency may support client adoption (an objective, model-derived trigger is more auditable than a self-reported claim), can reduce moral hazard under stable index–loss dependence, and makes pre-loss anticipatory payout conceptually feasible where forecast skill is confirmed. These claims remain empirically unvalidated in the present study and are offered as conditional propositions, not empirical conclusions.

Any dynamic or adaptive trigger formulations discussed elsewhere in the thesis are retained only as conceptual templates and are not part of the core operational design.

Beyond this validated core, the thesis also notes — but does not claim as a result — that the disagreement between the two best DM-confirmed models (the cross-model spread Δ) could serve as an optional, discretionary loading on the premium when models diverge. With no granular loss data

this signal has not been tested; it is offered only as an honest, optional add-on to the pricing layer, never as an input to the fixed trigger, and its formal form is left to future work (Section 6.4.1).

6.3 Limitations and Future Work

The contribution of this thesis should be understood as providing an evidence-based hazard and pricing layer, not a fully loss-calibrated insurance product. In the absence of insurance loss data, the proposed design is justified by statistical forecast skill (RMSE and DM tests) and by the economic logic of aligning premiums with seasonal risk via $Pr(I > \tau \mid \hat{\mu}, \sigma_f)$, rather than by demonstrated empirical reduction of basis risk at the policyholder level. Insurers can already benefit from this layer through forecast-conditioned premiums and forward-looking SCR signals, while loss-based trigger optimisation and indemnity-layer calibration are left explicitly to future work.

Primary limitation: No actual insurance loss data were used. The thesis builds a hazard module; the loss module — calibrating E3CL index values to actual agricultural or property losses — is the primary avenue for future work. ISTAT crop loss declarations for Emilia-Romagna paired with E3CL drought data would allow direct calibration of the trigger threshold against realised losses.

DM test power: With ~100 test observations per variable, the DM test has limited power to confirm moderate skill improvements (e.g., Ensemble 46% RMSE improvement at drought $h=1$ (RMSE 0.607 vs climatology 1.123) is non-significant). A longer observational record would resolve several borderline cases.

Single-station scope: Results are specific to Bologna (BO). Replication across the other 8 Emilia-Romagna E3CL stations would validate spatial generalisation and allow pooled products.

Hail mechanism: The hail skill source remains unconfirmed (4.3); SHAP feature-importance analysis is the priority next step.

Distribution assumption: The Gaussian exceedance probability approximation is valid only for drought. All other variables require GEV or empirical calibration for production use.

Future extensions: (1) Joint trigger design for compound drought-heat events; (2) Integration with NGFS forward-looking climate scenarios for trend-adjusted triggers; (3) SHAP-based model interpretability for EIOPA compliance; (4) Pseudo-replication using pre-2020 data as alternative hold-out periods to test DM result stability.

6.4 Conceptual Extensions and Future Work

This section collects two extensions of the forecast-informed pricing framework of Chapter 5. Neither is part of the operational pricing contribution; both are conceptual and identified as future research, the principal barrier in each case being the absence of granular insurance loss data.

6.4.1 Data-driven adaptive (moving) trigger — the lead future direction

The framework developed in this thesis fixes the contractual trigger at a transparent climatological percentile and lets the forecast move only the probability of crossing it. The natural next step is to let the threshold itself adapt to loss experience. A natural extension of this framework is the introduction of adaptive or moving trigger thresholds, where the payout threshold itself is learned from realised loss data using actuarial or machine-learning methods. This extension is not implemented in this thesis due to the absence of granular loss data, but it is identified as a promising direction for future research

in data-driven insurance design. No adaptive trigger is estimated or back test in this thesis, and no numerical example is provided, because the required loss data are not available.

The two designs differ in kind, not degree. The fixed-trigger model used here keeps the contract transparent and stable, lets the forecast change only the price, and is deployable now — it is a pricing innovation on an unchanged contract structure. The adaptive-trigger model would let the threshold itself respond to realised losses, requiring loss data, calibration, and an actuarial or machine-learning feedback loop — it is a structural redesign of the contract, less transparent and not yet operational. The threshold there would be learned from losses, not anchored to the forecast mean: anchoring the trigger to the forecast would hold the activation probability roughly constant and remove the premium's responsiveness, so the loss-driven formulation is the appropriate one. Once an insurer's loss series is available, the same forecasting models evaluated in this thesis could be used to learn the threshold, closing the loop between forecast skill and contract design. Once insurer loss series become available, the same DM-validated forecasting pipeline developed in this thesis could be used to learn loss-optimal thresholds, turning the adaptive trigger into a separate actuarial–ML research project beyond the scope of the present thesis.

The remainder of this section presents a moving-threshold form only as a conceptual extension for future actuarial calibration; it is NOT the design implemented or priced in this thesis, which fixes the contractual trigger and lets the forecast move only the conditional crossing probability and hence the premium. In that conceptual extension — feasible only once granular loss data are available to calibrate the stringency level k — the moving trigger threshold at time t , for horizon h , and stringency level k , would be:

$$\tau_k(t) = \hat{\mu}_{t+h} + k \cdot \Delta_{t+h}$$

where $\hat{\mu}_{t+h}$ is the DM-confirmed model's point forecast at horizon h , $k \in \{0.5, 1.0, 1.5, 2.0\}$ is the trigger stringency (controlling false-positive/false-negative trade-off), and Δ_{t+h} is the cross-model spread of the two best models' forecasts at that horizon:

$$\Delta_{t+h} = \sqrt{\left[\frac{1}{2} \cdot (\hat{y}_{1,t+h} - \hat{y}_{2,t+h})^2 \right]}$$

where $\hat{y}_1$ and $\hat{y}_2$ are the forecasts of the 2 best models. This is a 2-model sample standard deviation. Critically, this Δ is not a historical standard deviation — it is a forward-looking, current-season measure of how much the best models disagree about what will happen at horizon h . When models agree, Δ is small and the threshold is tight. When models disagree, Δ is wide and the threshold signals higher uncertainty, which should be reflected in a higher premium or a wider payout range.

6.4.2 Hybrid climate-risk financing architecture

The hybrid three-layer architecture is presented solely as a conceptual future-work extension and is not part of the core operational contribution. The design combines: (i) an anticipatory payment triggered when the forecast-conditioned exceedance probability exceeds a warning threshold; (ii) an automatic parametric payout if the realised index exceeds the fixed trigger; and (iii) a residual indemnity layer covering losses not captured by the index. Unlike the Chapter 5 framework, where forecasts only affect pricing, the hybrid structure also provides pre-season liquidity to policyholders. Forecast-error exposure is limited because only the anticipatory layer depends on forecast information. Although the architecture may reduce basis risk through the indemnity layer, it is not

calibrated or validated in this thesis due to the absence of insurance loss data. Joint pricing, SCR recalibration, and optimal layer allocation are therefore left to future research.

6.5 Closing Remark

In summary, this thesis has one principal operational contribution — the forecast-informed pricing mechanism on a fixed climatological trigger — while adaptive triggers and hybrid payout structures are intentionally confined to conceptual future-work status due to the absence of loss data.

Multi-station replication and crop-specific basis-risk estimation, of the kind increasingly emphasised in recent agricultural-finance and sustainability work (Parhi & Panda, 2026), are essential next steps before the framework can support real underwriting decisions. Future work should also include a simple historical back-test of trigger activation over 1981–2025 under both the static and dynamic designs, to show how often each rule would have fired and whether the resulting activation frequencies are actuarially sensible.

As green finance regulation continues to tighten, the demand for statistically validated and reproducible climate triggers is likely to increase. The EU Green Bond Standard, the Sustainable Finance Disclosure Regulation (SFDR), and EIOPA’s climate-related Own Risk and Solvency Assessment (ORSA) guidance all point toward greater transparency, stronger disclosure, and more disciplined treatment of climate-related risk in financial products. In that regulatory environment, a trigger framework cannot rely on intuition or arbitrary historical thresholds alone; it must demonstrate that its trigger logic is both observable and statistically defensible.

The contribution of this thesis is precisely to offer such a structure: an open, DM-validated forecasting layer that can support forecast-conditioned pricing and parametric index design where climate uncertainty matters for pricing, capital, or payout logic. The insurance case is the primary demonstration — one in which the DM gate ensures that only statistically credible forecasts feed the pricing calculation, and the cross-model spread Δ provides a transparent, non-parametric uncertainty premium. The broader message is that green finance increasingly needs the same kind of contribution this thesis provides: climate signals that can be audited, replicated, and defended before regulators, investors, and counterparties. The methodological contribution — a forecast-governance framework grounded in the Diebold–Mariano test and operationalised in this thesis through the fixed-threshold, forecast-conditioned pricing variant and presented conceptually through the dynamic trigger rule $\tau_k(t) = \hat{\mu}_{t+h} + k \cdot \Delta_{t+h}$ as a framework for future deployment — is a step toward providing that standard for the Greening Energy Market and Finance regulatory context in which this thesis is situated.

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Appendix A: Complete DM Test Results by Variable

Full DM test tables are presented below for all six variables across all four horizons ($h = 1, 3, 6, 12$). Significance tiers: *** $p < 0.001$ (strong family-wise evidence); ** $p < 0.01$; * $p < 0.05$ (suggestive evidence). Consistent with the three-tier framework in Section 3.4. Table A.1 reports the full Diebold–Mariano statistics and p-values for every drought model at all four horizons and Table A.2 reports the corresponding DM statistics for hail.

A.1 Drought — DM Test Results (Full)

Model	h1 DM	h1 p	h3 DM	h3 p	h6 DM	h6 p	h12 DM / p
Seasonal Naive	-2.40 *	0.020	-3.47 **	0.001	-2.52 *	0.015	-4.48** / <.001
ETS	0.61	0.542	-2.32 *	0.024	-3.46**	0.001	-4.44** / <.001
RidgeSeasonalDecomp	-0.00	0.999	-3.33**	0.002	-3.49**	0.001	-2.55 * / 0.014
SARIMA	1.58	0.121	-0.02	0.981	-0.90	0.373	-1.84 / 0.072
XGBoost_STL	1.16	0.254	-1.07	0.291	1.01	0.317	0.39 / 0.697
PyTorchLSTM	1.57	0.137	-1.09	0.293	-0.09	0.926	0.93 / 0.364
Prophet	0.67	0.503	-0.23	0.818	-0.01	0.995	2.07* / 0.044
Ensemble	1.12	0.278	0.46	0.650	0.19	0.855	3.25** / 0.002

Table A.1. Drought DM test statistics vs Monthly Climatology. * $p < 0.05$; ** $p < 0.01$. Positive = model better than climatology.

A.2 Hail — DM Test Results (Full)

Model	h1 DM	h1 p	h3 DM	h3 p	h6 DM	h6 p	h12 DM / p
Seasonal Naive	-2.18 *	0.034	-2.60 *	0.012	-1.66	0.103	-2.26* / 0.028
RidgeSeasonalDecomp	-2.20 *	0.033	-0.88	0.384	-0.15	0.885	0.19 / 0.851
ETS	0.46	0.646	1.91	0.061	0.84	0.404	1.66 / 0.103
Prophet	-0.71	0.480	0.31	0.761	2.61*	0.012	1.12 / 0.267
PyTorchLSTM	-0.07	0.948	1.23	0.237	0.54	0.599	-8.12** / <.001
Ensemble	-0.28	0.786	2.78*	0.013	1.64	0.107	4.74*** / .0002

Table A.2. Hail DM test statistics vs Monthly Climatology. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

A.3 DM Test Summary for Remaining Variables

The Table A.3. provides DM test results for forest fire, extreme temperature, extreme precipitation, and extreme wind. Positive DM values indicate the model outperforms Monthly Climatology; asterisks denote significance tiers per the three-tier framework (Section 3.4).

Hazard index	Horizon (h)	Model	DM statistic	p-value	Interpretation
Forest fire	h=3	XGBoost_STL	+3.423*	0.019	Significantly better
Forest fire	h=1,3,6	Other models	—	—	Not significant
Forest fire	h=12	All models	—	—	Not available (numerical instability)
Forest fire	h=1	Seasonal Naive	−2.18*	0.033	Significantly worse
Extreme max temperature	h=6	SARIMA	+2.211*	0.032	Significantly better
Extreme max temperature	h=6	XGBoost_STL	−2.446*	0.018	Significantly worse
Extreme max temperature	h=12	XGBoost_STL	−2.064*	0.044	Significantly worse
Extreme max temperature	h=1,3	All models	—	—	Not significant
Extreme precipitation	all h	All models	—	—	No positive significant results
Extreme precipitation	h=12	LightGBM	−2.796**	0.006	Significantly worse
Extreme precipitation	h=12	PyTorch LSTM	−3.330**	0.002	Significantly worse
Extreme wind	h=1	XGBoost_MV_Spatial	+2.088*	0.042	Spatial coherence (not temporal skill)
Extreme wind	h=3	BaggedXGB	+2.506*	0.016	Variance reduction effect
Extreme wind	h=12	LightGBM	−2.193*	0.033	Significantly worse

Table A.3. DM test results for Extreme wind, Extreme precipitation, Forest Fire, Extreme temperature

Appendix B: Full RMSE and R² Tables by Variable

Table B.1 gives drought RMSE and R² for every model across all horizons (best per column in bold) and Table B.2 gives the same breakdown for hail.

B.1 Drought — RMSE across All Horizons

Model	h1 RMSE	h1 R ²	h3 RMSE	h6 RMSE	h12 RMSE	h12 R ²	DM h12
Monthly Climatology	1.123	-0.095	1.092	1.138	1.225	-0.067	—
SARIMA	0.918	0.267	1.093	1.172	1.253	-0.116	-1.84
ETS	1.031	0.076	1.357	1.638	1.746	-1.165	-4.44 **
Prophet	1.101	-0.054	1.099	1.138	1.168	0.031	2.07 *
XGBoost_STL	1.078	-0.010	1.133	1.114	1.212	-0.044	0.39
PyTorchLSTM	0.599	0.313	0.794	0.882	1.266	-0.030	0.93
Ensemble	0.607	0.294	0.734	0.872	1.131	0.091	3.25 **

Table B.1. Drought RMSE and R² across all horizons. Bold = DM-confirmed best at h=12.

B.2 Hail — RMSE across All Horizons

Model	h1 RMSE	h1 R ²	h3 RMSE	h6 RMSE	h12 RMSE	h12 R ²	DM h12
Monthly Climatology	1.005	-0.017	1.051	1.141	1.072	-0.043	—
SARIMA	1.001	-0.009	1.063	1.130	1.060	-0.021	1.74
ETS	1.001	-0.008	1.018	1.124	1.051	-0.003	1.66
Prophet	1.034	-0.076	1.043	1.031	1.024	0.048	1.12
PyTorchLSTM	0.613	-0.001	1.014	1.483	1.003	-0.173	-8.12 ***
Ensemble	0.620	-0.024	1.027	1.096	0.898	0.059	4.74 ***

Table B.2. Hail RMSE and R² across all horizons. Ensemble h=12 DM=4.743*** is the strongest positive result in the study. *** $p < 0.001$ (robust under Bonferroni correction); ** $p < 0.01$; * $p < 0.05$. PyTorchLSTM h=6 DM=-8.121 is the strongest negative.

B.3 RMSE Summary — Remaining Four Variables

Table B.3 provides RMSE at the benchmark (Monthly Climatology) and best-model RMSE at each horizon for the four remaining variables. Cells marked with asterisk indicate the model at that horizon achieved DM-confirmed positive skill.

Variable	Horizon	Climatology RMSE	Best Model	Best RMSE	Key Finding
Forest Fire	h=1	1.675	Ensemble	1.162	Best performance among tested models
Forest Fire	h=3	1.611	XGBoost_STL*	1.397	DM-confirmed improvement
Forest Fire	h=6	1.536	Ensemble	1.476	Marginal improvement
Forest Fire	h=12	0.714	Ensemble	0.636	Improvement observed; DM unavailable due to numerical instability
Extreme Max Temperature	h=1	1.740	SARIMA	1.744	No meaningful improvement over climatology
Extreme Max Temperature	h=3	1.802	SARIMA	1.802	Equivalent to climatology
Extreme Max Temperature	h=6	1.812	SARIMA*	1.604	Only DM-confirmed positive result
Extreme Max Temperature	h=12	1.833	Prophet	1.664	Improvement not DM-confirmed
Extreme Precipitation	h=1	2.303	ETS	0.926	Spuriously low RMSE from near-constant predictions; not considered valid skill
Extreme Precipitation	h=3	2.488	—	—	No DM-confirmed positive skill
Extreme Precipitation	h=6	2.496	—	—	No DM-confirmed positive skill
Extreme Precipitation	h=12	2.703	—	—	No DM-confirmed positive skill; LightGBM RMSE = 3.842 and PyTorchLSTM RMSE = 3.119 significantly worse
Extreme Wind	h=1	0.856	XGBoost_MV_Spatial*	0.717	DM-confirmed via spatial coherence
Extreme Wind	h=3	0.908	BaggedXGB*	0.824	DM-confirmed via variance reduction
Extreme Wind	h=6	0.918	—	—	No confirmed forecast skill
Extreme Wind	h=12	0.904	—	—	No confirmed forecast skill; LightGBM RMSE = 1.012 significantly worse

Table B. 3. RMSE horizon comparison for Extreme wind, Extreme precipitation, Forest Fire, Extreme temperature

Appendix C: Additional Figure Positions

C.1 Drought — Additional Figures

Figures C.1–C.6 provide the per-horizon model comparisons (RMSE, MAE, R^2 , SMAPE) and walk-forward plots for drought at $h = 1, 3, 6$, and 12 .

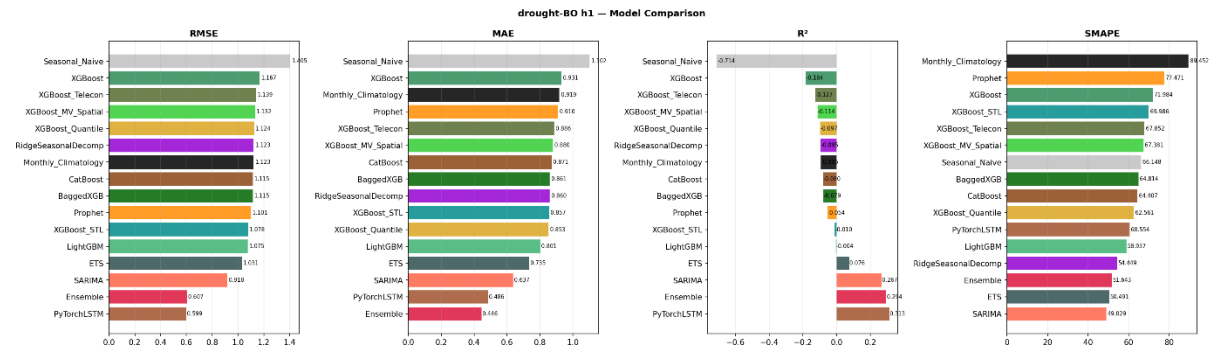


Figure C.1. Model comparison at $h=1$ for drought: RMSE, MAE, R^2 , SMAPE bar charts. PyTorchLSTM and Ensemble lead on RMSE and R^2 .

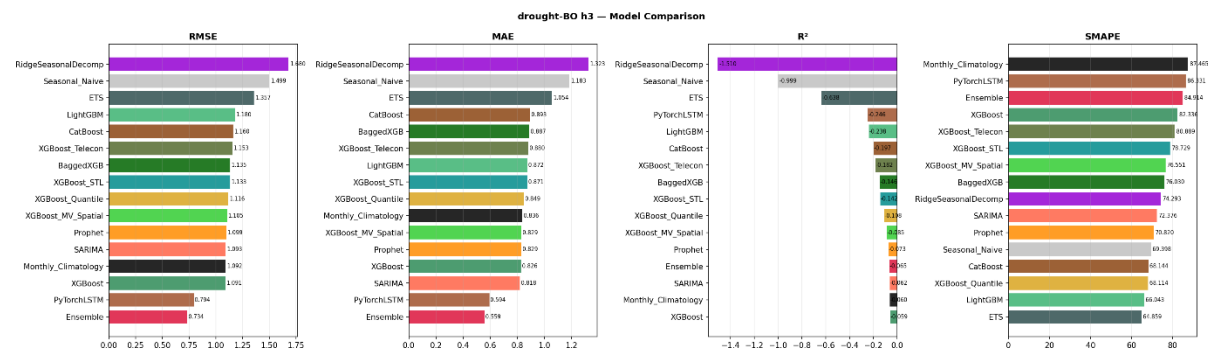


Figure C.2. Model comparison at $h=3$ for drought: RMSE, MAE, R^2 , SMAPE bar charts.

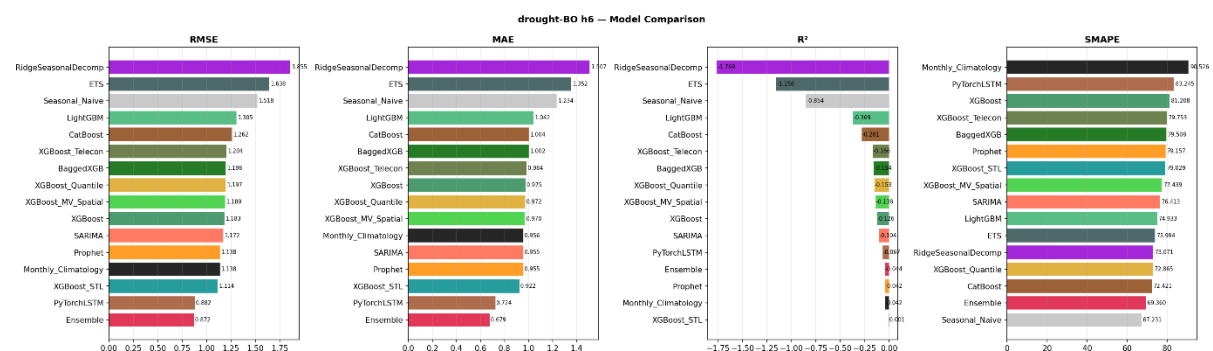


Figure C.3. Model comparison at $h=6$ for drought: RMSE, MAE, R^2 , SMAPE bar charts.

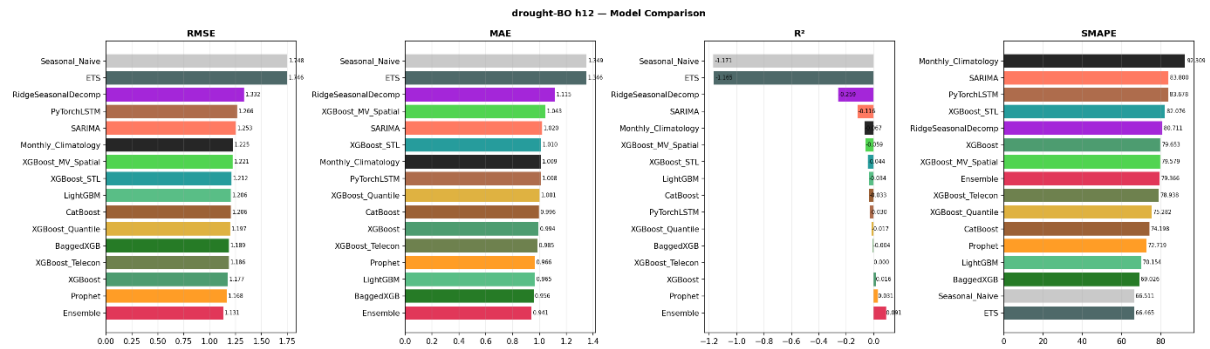


Figure C.4. Model comparison at $h=12$ for drought: Ensemble and Prophet confirmed superior; ETS confirmed worst.

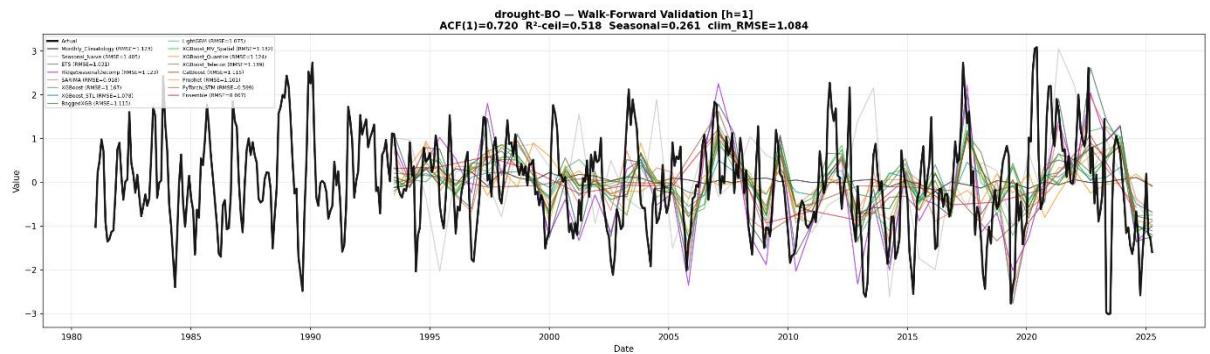


Figure C.5. Walk-forward validation at $h=1$ for drought (1981–2025). Shows PyTorchLSTM and Ensemble tracking the series closely.

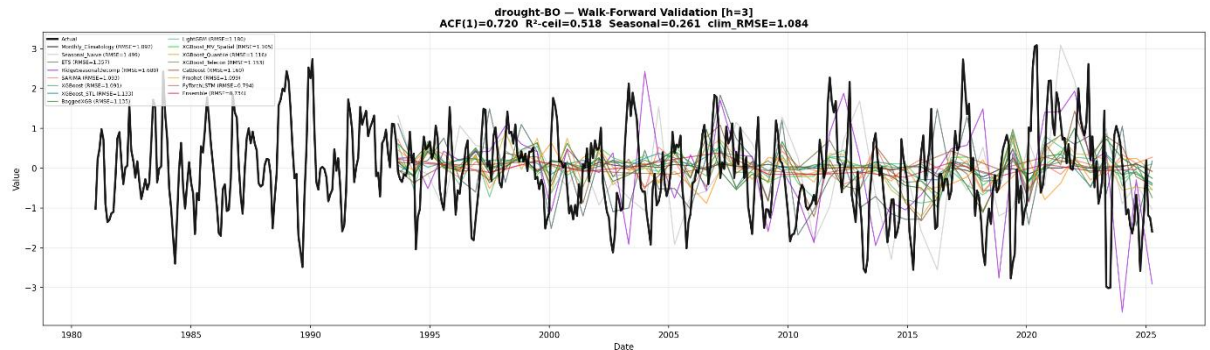


Figure C.6. Walk-forward validation at $h=3$ for drought.

C.2 Hail — Additional Figures

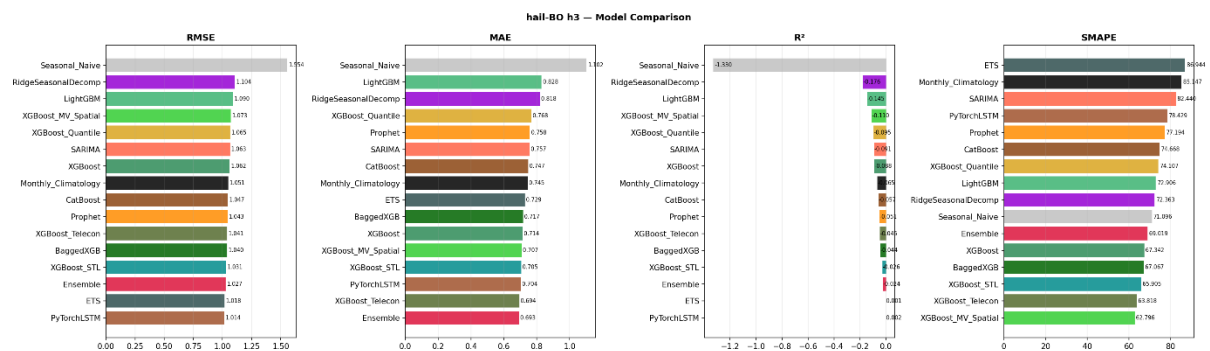


Figure C.7. Model comparison at $h=3$ for hail: Ensemble DM-confirmed (DM=2.783).

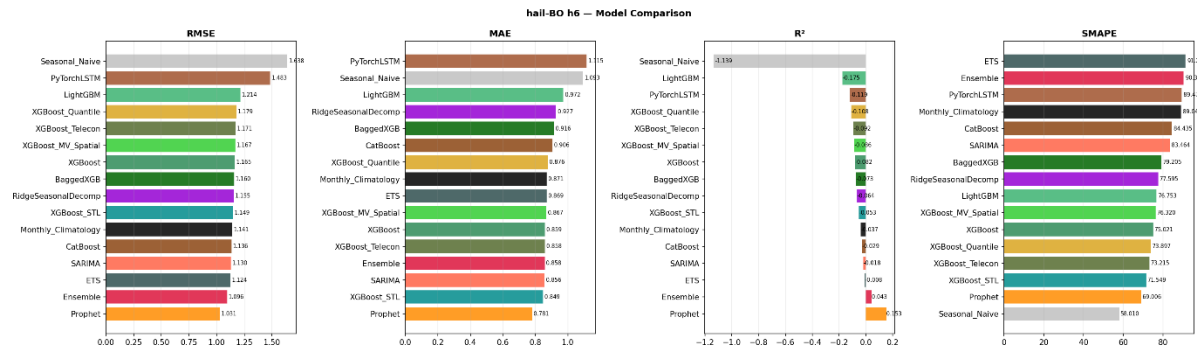


Figure C.8. Model comparison at $h=6$ for hail: Prophet confirmed ($DM=2.607$, $p=0.012$); PyTorchLSTM non-significant ($DM=0.54$, $p=0.599$).

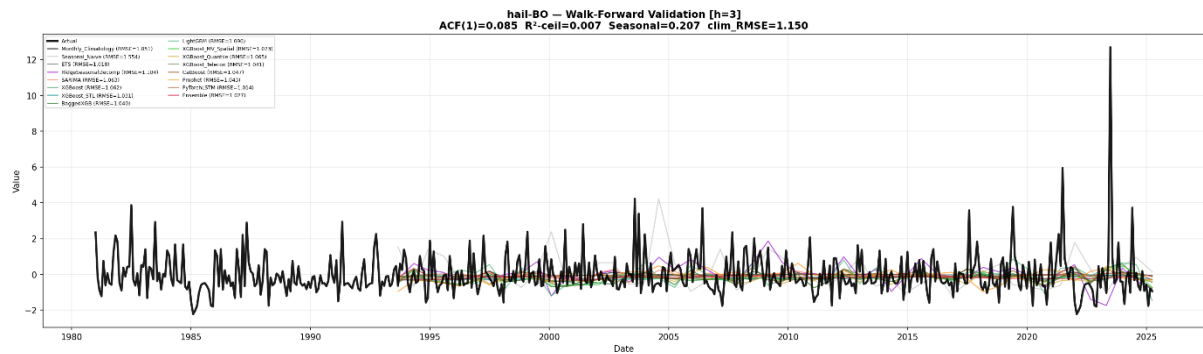


Figure C.9. Walk-forward validation at $h=3$ for hail.

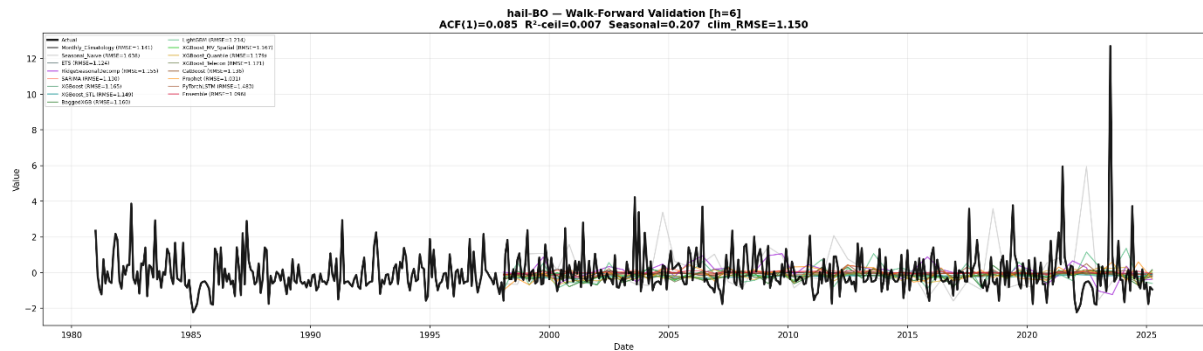


Figure C.10. Walk-forward validation at $h=6$ for hail. Shows PyTorchLSTM drift.

C.3 Forest Fire — Additional Figures

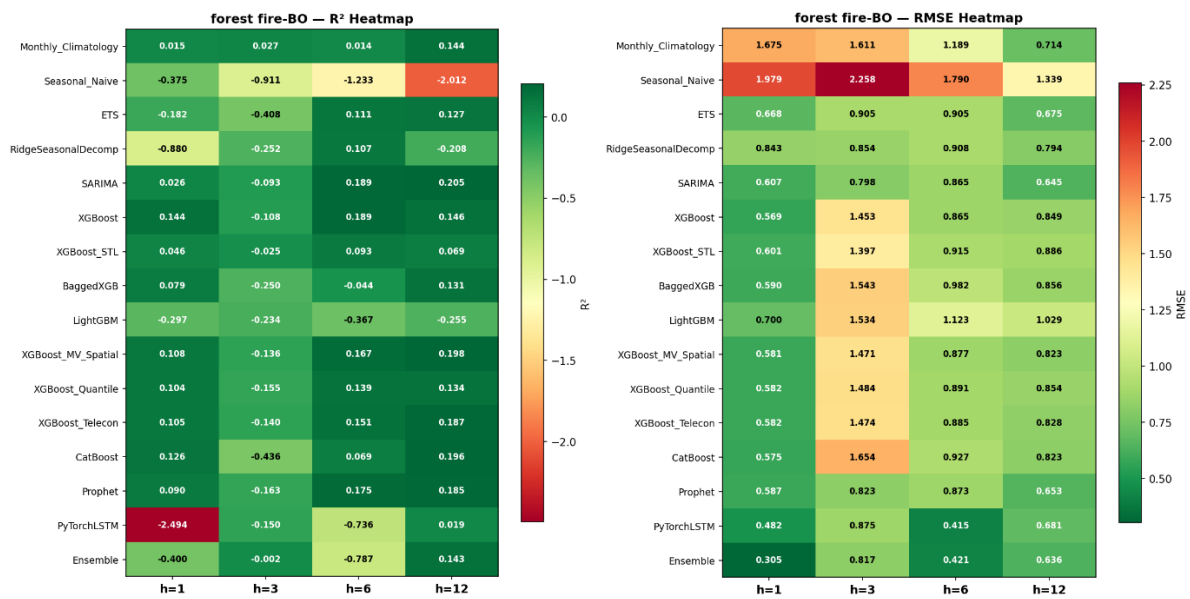


Figure C.11. RMSE and R^2 heatmaps for forest fire. XGBoost_STL confirmed at h=3 only.

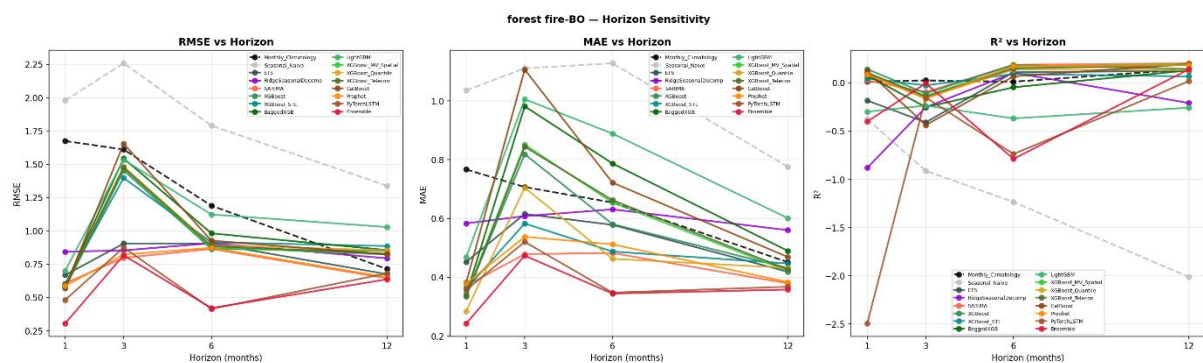


Figure C.12. Horizon sensitivity for forest fire. Note the unusual h=12 climatology RMSE drop (December exclusion).

C.4 Extreme Temperature — Additional Figures

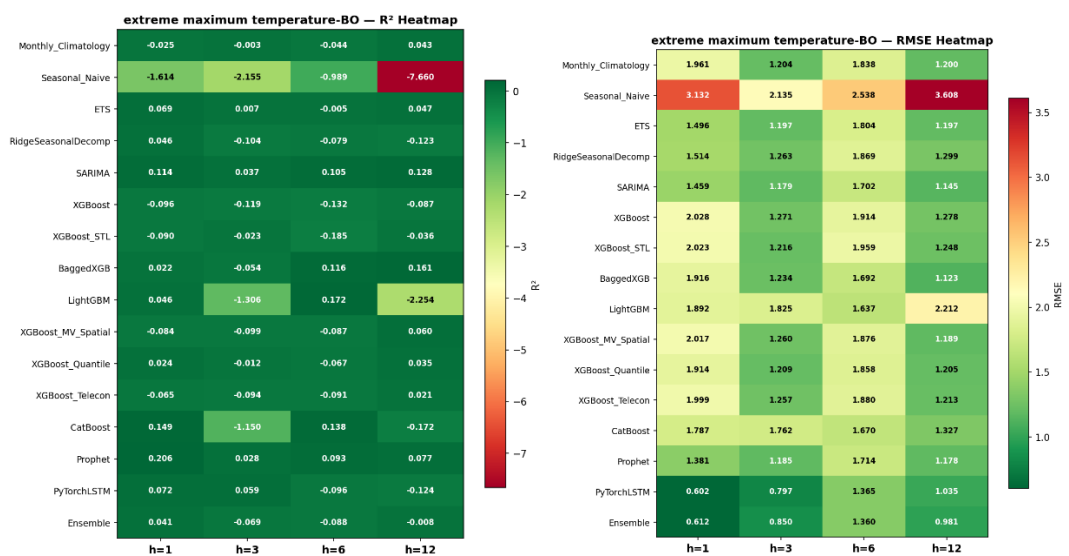


Figure C.13. RMSE and R^2 heatmaps for extreme T. SARIMA confirmed at $h=6$; XGBoost_STL confirmed worse at $h=6$ and 12.

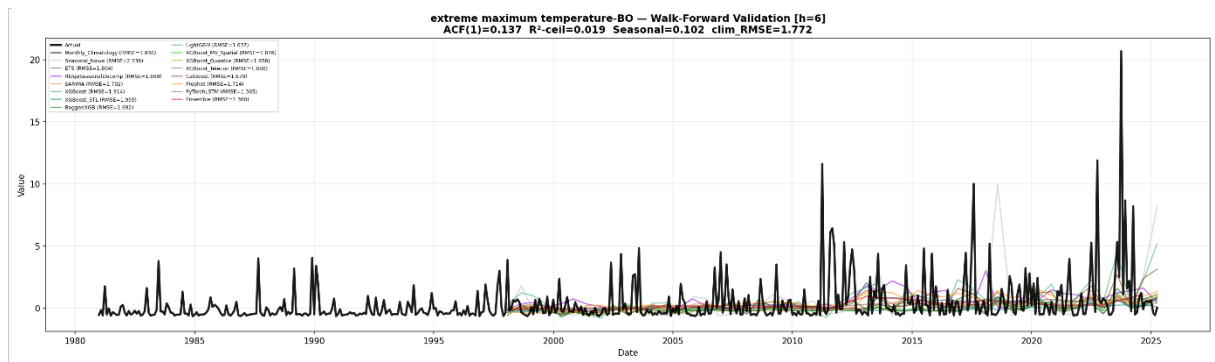


Figure C.14. Walk-forward validation at $h=6$ for extreme temperature shows SARIMA's skill advantage.

C.5 Extreme Wind — Additional Figures

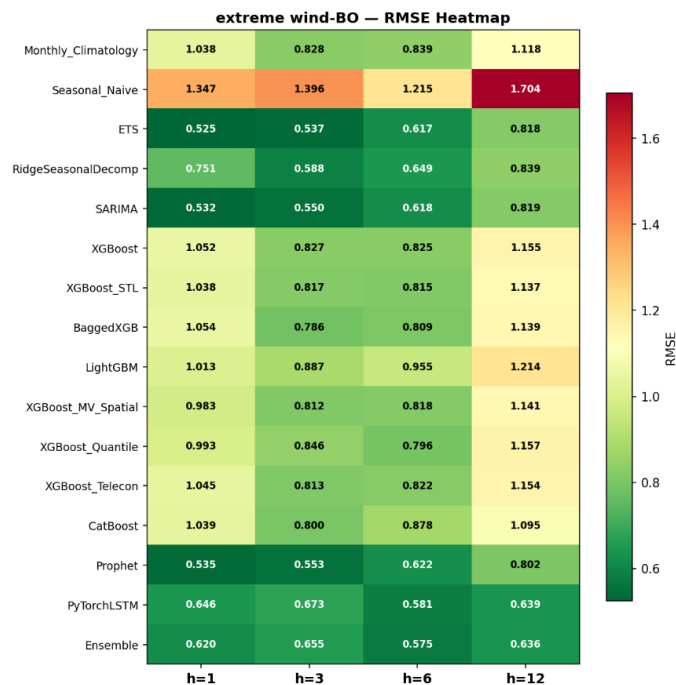


Figure C.15. RMSE heatmap for extreme wind. XGBoost_MV_Spatial confirmed at $h=1$ (spatial coherence mechanism). LightGBM confirmed worse at $h=12$

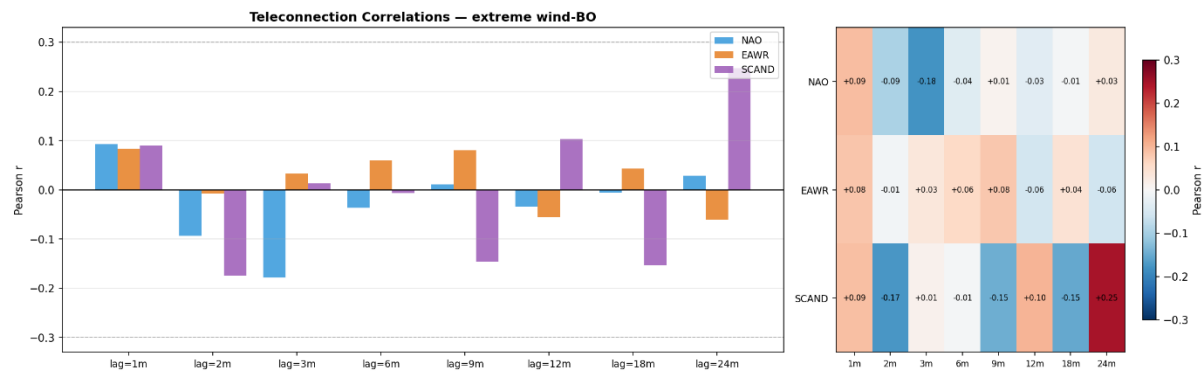


Figure C.16. Teleconnection correlations for extreme wind: NAO and EAWR at lag=24 show long-range correlations (+0.25) but no actionable short-lag signal.

Appendix D: Illustrative Premium Arithmetic — Two Complementary Scenarios

This appendix provides the full mechanical arithmetic underlying the premium sensitivity example introduced in Section 5.3. It reproduces the same $h = 12$ drought scenario used in the main text — the same forecast setup, the same forecast pair, the same $\hat{\mu}$, and the same trigger — so that every number in Table 5.2 can be checked step by step. No additional scenarios or horizons are introduced here; the $h = 6$ illustration in Section 5.3.1 is self-contained and does not require separate appendix arithmetic. All figures below are illustrative; see 5.1.1 for the full scope of this framework.

D.1 Shared Assumptions

The calculations below use the same assumptions as Section 5.3.

Shared Assumptions (both scenarios)

Parameter	Value	Source / Note
Notional payout P	€1,000,000	Mid-scale agricultural cooperative (Swiss Re Sigma, 2024)
Loading factor λ	0.20 / 0.30 / 0.40	Catastrophe reinsurance ROL range (Geneva Association, 2024)
Distribution	Gaussian	Empirically valid for drought only (JB $p = 0.84$)
Historical mean μ_{hist}	-0.049	Full-sample E3CL drought index, Bologna
Historical std σ_{hist}	1.088	Full-sample E3CL drought index, Bologna
Contractual trigger τ	+1.37	90th percentile (P90) of the historical drought index; fixed throughout.
Forecast horizon h	12 months	Stress-test horizon; DM significance confirmed
Forecast pair	Ensemble (+0.72), Prophet (+0.60)	DM-confirmed models at $h = 12$
Forecast-error spread σ_f	1.131	Walk-forward Ensemble RMSE at $h = 12$ (Appendix B), used as a practical estimate of σ_f under approximately unbiased errors; conditional spread in the dynamic probability.
Forecast mean $\hat{\mu}$	+0.66	$\frac{1}{2}(0.72 + 0.60)$; same value as Section 5.3
Cross-model spread $\Delta_{\{t+12\}}$	0.085	$\sqrt{[\frac{1}{2}(0.72 - 0.60)^2]}$; Model-disagreement diagnostic / optional loading; not used in the trigger or the conditional spread.
Index-to-loss ρ	1.0 (assumed)	Upper bound; true R^2 estimated at 0.30–0.60 in comparable studies

Note on the two σ symbols: $\sigma_{hist} = 1.088$ is the historical standard deviation of the drought index itself, used in all exceedance probability calculations below. $\Delta_{\{t+12\}} = 0.085$ is the cross-model forecast spread, which serves as a forecast-uncertainty diagnostic. These are different quantities and must not be confused.

D.2 Static Design

The static design uses historical climatology only, with no forecast information.

Step 1. The trigger $\tau = +1.37$ is the empirical 90th percentile of the Bologna drought index, so by construction it is exceeded in 10.0% of historical months (no Gaussian assumption needed for the unconditional baseline).

Step 2. $E[L]_{static} = 0.100 \times €1,000,000 = €100,000$

Step 3. Premiums: $\lambda=0.20 \rightarrow \text{€}120,000$; $\lambda = 0.30 \rightarrow \text{€}130,000$; $\lambda = 0.40 \rightarrow \text{€}140,000$

D.3 Forecast-Informed $h = 12$ Case (Elevated-Drought Scenario)

This is the forecast-informed design used in Section 5.3 and Table 5.2.

Step 1. $\hat{\mu}_{\{t+12\}} = \frac{1}{2}(0.72 + 0.60) = +0.66$ (elevated-drought scenario).

Step 2. $z_{dyn} = \frac{\tau - \hat{\mu}}{\sigma_f} = \frac{1.37 - 0.66}{1.131} = 0.624$ (Note: the conditional spread is $\sigma_f = 1.131$, not σ_{hist} .)

Step 3. $\Pr(I > \tau \mid \hat{\mu} = +0.66, \sigma_f = 1.131) = 1 - \Phi(0.624) \approx 26.6\%$

Step 4. $E[L]_{dynamic} = 0.266 \times \text{€}1,000,000 = \text{€}266,200$

Step 5. Premiums: $\lambda=0.20 \rightarrow \text{€}319,500$; $\lambda=0.30 \rightarrow \text{€}346,100$; $\lambda=0.40 \rightarrow \text{€}372,700$

The crossing probability rises from 10.0% (static) to 26.6% (forecast-informed) purely through the elevated forecast mean; the trigger $\tau = +1.37$ does not move.

D.4 Summary Table

The table below reproduces Table 5.2 from Section 5.3, with the intermediate steps documented in D.2–D.3 above.

Design	$E[L]$	Premium ($\lambda = 0.20$)	Premium ($\lambda = 0.30$)	Premium ($\lambda = 0.40$)
Static ($\mu_{hist} = -0.049$)	€100,000	€120,000	€130,000	€140,000
Forecast-informed ($\hat{\mu} = +0.66$, $h = 12$)	€266,200	€319,500	€346,100	€372,700
Difference (Forecast-informed – Static)	+€166,200	+€199,500	+€216,100	+€232,700

Table D.1. at an index-to-loss $R^2 \approx 0.45$, effective $E[L] \approx 0.45 \times \text{€}266,200 = \text{€}119,800$ forecast-informed vs $0.45 \times \text{€}100,000 = \text{€}45,000$ static — still a meaningful difference. Calibrating the true index-to-loss relationship remains the primary future work.